Lecture Notes in Microeconomic Theory

Mechanism and Information Design

Independent Study · Fall 2026

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Preface

These lecture notes are the result of my independent study of microeconomic theory, with a focus on mechanism design and information design. They bring together results from the published literature, original proofs developed as part of my study, and questions for further research.

Theorems, propositions, and lemmas drawn from the literature are attributed to their original sources. Proofs developed independently are explicitly identified as such. Each part concludes with questions intended to encourage further exploration of the topics covered.

Any errors or omissions are my own.

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CHAPTER **1**

Mechanism Design

1.1 Why Mechanism Design?

1.1.1 Motivating Example: Selling an Indivisible Good

Suppose you own an indivisible good, such as a piece of craft, a piece of land or a painting. You want to sell it and earn revenue. Your goal is to design a selling mechanism that maximizes your profit. But there is a problem: you do not know the buyer's true valuation v. The buyer knows how much the good is worth to them, but they may have an incentive to misreport their valuation to obtain a lower price.

This raises some important questions:

1. How should you set the price?
2. How can you encourage truthful reporting? If buyers report their valuations, how can you ensure they have no incentive to lie?
3. If several buyers compete for it, who should receive the good?
4. Can you maximize revenue while ensuring the buyer has incentives to participate?

The central idea of mechanism design: rather than simply accepting the market (or game in general) as given, you *design the rules* of the interaction to achieve a desired outcome despite participants having private information and their own incentives. The challenge is to design rules so that the seller's objectives are achieved while accounting for the buyer's incentives.

Before building the general model, it is useful to see how far the simplest selling rule takes us.

Example 1.1 (Posted price with one buyer). A seller faces one buyer whose value v is drawn from the uniform distribution on $[0, 1]$. The seller posts a take-it-or-leave-it price p. The buyer buys if and only if $v \geq p$, which happens with probability $1 - p$. Expected revenue is

$$R(p) = p\,(1 - p),$$

which is maximized at $p^* = 1/2$, giving revenue $R(p^*) = 1/4$.
Two features are worth noting.

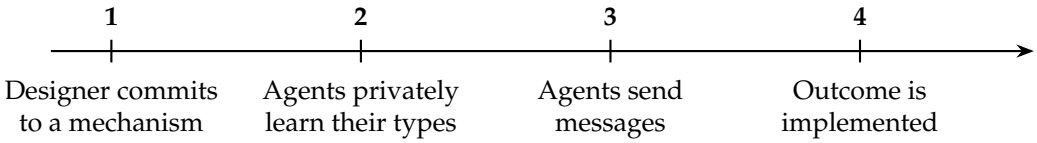
(a) The buyer has no reason to lie: they never report anything, they simply accept or reject. Truth-telling is built into the rule.

(b) The outcome is *inefficient*: whenever $v < 1/2$, the good is not sold even though the buyer values it more than the seller (who values it at 0). The seller deliberately sacrifices some surplus to extract more of it.

We will see in Section 1.4 that, remarkably, no mechanism of any kind can do better than this posted price.

1.1.2 The Economic Environment and Model Setup

The interaction we study has the following timing.



The essential assumption is **commitment**: the designer announces the rules first and cannot change them after observing the messages. Without commitment, the designer would be tempted to use what she learns against the agents, and the agents, anticipating this, would not report truthfully.

The designer may be a seller (who cares about revenue), a government (who cares about efficiency or equity), a platform, or an employer. What makes the problem one of mechanism design is that the information needed to reach the designer's goal is held by the agents, not the designer.

1.1.3 Agents, Types, Preferences, and Feasible Outcomes

Definition 1.2 (Environment). An **environment** consists of:

(i) a finite set of **agents** $N = \{1, \dots, n\}$;

(ii) for each agent i, a set of **types** Θ_i; we write $\Theta = \Theta_1 \times \cdots \times \Theta_n$ and $\theta = (\theta_i, \theta_{-i}) \in \Theta$;

(iii) a common **prior** F on Θ, from which types are drawn;

(iv) a set of **outcomes** X;

(v) for each agent i, a **utility function** $u_i : X \times \Theta \to \mathbb{R}$.

Agent i privately observes θ_i. The prior F, the sets Θ_i and X, and the utility functions are common knowledge.

The type θ_i summarizes everything agent i knows that others do not: a valuation, a cost, a skill level, a preference ranking. When $u_i(x, \theta)$ depends only on θ_i, we say values are **private**; otherwise they are **interdependent**.

Most of these notes use a special but very important class of environments.

Definition 1.3 (Quasilinear environment). An environment is **quasilinear** if each outcome has the form $x = (k, t)$, where $k \in K$ is an **allocation** (a "physical" decision) and $t = (t_1, \dots, t_n) \in \mathbb{R}^n$ is a vector of **payments** from the agents to the designer, and

$$u_i(k, t, \theta) = v_i(k, \theta_i) - t_i.$$

Example 1.4 (Single-object auction). One indivisible good is to be allocated among n buyers. An allocation is a vector $q = (q_1, \dots, q_n)$ with $q_i \geq 0$ and $\sum_i q_i \leq 1$, where q_i is the probability that buyer i receives the good. Buyer i has value θ_i and utility

$$u_i(q, t, \theta) = \theta_i q_i - t_i.$$

We assume throughout this case that types are independent, $\theta_i \sim F_i$ on $[\underline{\theta}, \bar{\theta}]$, with a density $f_i > 0$. This is the **independent private values** (IPV) model.

The designer's goal is described by a **social choice function** $f : \Theta \to X$, which specifies the outcome she would like to implement in each state of the world. For example, *efficiency* in the auction setting means giving the good to the buyer with the highest value.

The difficulty is that f depends on θ, which the designer does not observe.

1.2 Mechanisms and Incentive Compatibility

1.2.1 Direct and Indirect Mechanisms

Definition 1.5 (Mechanism). A **mechanism** $\Gamma = (M_1, \dots, M_n, g)$ consists of a set of **messages** M_i for each agent and an **outcome function** $g : M \to X$, where $M = M_1 \times \cdots \times M_n$. A mechanism is **direct** if $M_i = \Theta_i$ for every i: each agent is simply asked to report a type. A mechanism that is not direct is called **indirect**.

A mechanism together with the environment defines a Bayesian game. A **strategy** for agent i is a map $s_i : \Theta_i \to M_i$.

Example 1.6 (Direct and indirect selling rules).

(a) A **first-price sealed-bid auction** is indirect: messages are bids $b_i \geq 0$, the highest bidder wins and pays her own bid. Bids are not types, and in equilibrium buyers bid less than their values.

(b) A **second-price sealed-bid auction** can be seen as a direct mechanism: each buyer reports a value, the highest report wins and pays the second-highest report.

(c) An ascending (English) auction is an indirect, dynamic mechanism.

Direct mechanisms are easier to analyze, because there is a natural candidate for equilibrium: everyone tells the truth. A direct mechanism with outcome function f *is* a social choice function; we will use the same letter for both.

1.2.2 Dominant-Strategy Incentive Compatibility

Definition 1.7 (DSIC). A direct mechanism f is **dominant-strategy incentive compatible** (DSIC, or **strategy-proof**) if for every agent i, all $\theta_i, \theta_i' \in \Theta_i$ and all $\theta_{-i} \in \Theta_{-i}$,

$$u_i\big(f(\theta_i, \theta_{-i}), \theta_i\big) \;\geq\; u_i\big(f(\theta_i', \theta_{-i}), \theta_i\big). \tag{1.1}$$

Interpretation. Truth-telling is optimal *whatever the other agents report*. An agent does not need to know the prior, or to believe that others are truthful. This is the most robust form of incentive compatibility.

Proposition 1.8 (Vickrey, 1961). *In the single-object environment, the second-price auction is DSIC.*

Proof (standard; see Krishna, 2009, Ch. 2). Fix buyer i with value θ_i and the others' reports, and let $b = \max_{j \neq i} \theta_j$ be the highest competing report. If i reports r, she wins when $r > b$ and pays b, so her utility is $\theta_i - b$ if $r > b$ and 0 if $r < b$ (ties have probability zero and can be ignored).

- If $\theta_i > b$, any report $r > b$ (including $r = \theta_i$) gives the best possible utility $\theta_i - b > 0$.
- If $\theta_i < b$, any report $r < b$ (including $r = \theta_i$) gives the best possible utility 0; winning would give $\theta_i - b < 0$.

In both cases reporting $r = \theta_i$ is a best response, whatever b is. □

1.2.3 Bayesian Incentive Compatibility

Definition 1.9 (BIC). A direct mechanism f is **Bayesian incentive compatible** (BIC) if for every agent i and all $\theta_i, \theta_i' \in \Theta_i$,

$$\mathbb{E}_{\theta_{-i}}\big[u_i\big(f(\theta_i, \theta_{-i}), \theta_i\big) \,\big|\, \theta_i\big] \;\geq\; \mathbb{E}_{\theta_{-i}}\big[u_i\big(f(\theta_i', \theta_{-i}), \theta_i\big) \,\big|\, \theta_i\big]\,. \tag{1.2}$$

That is, truth-telling is a Bayes–Nash equilibrium of the direct mechanism.

Taking expectations of (1.1) shows immediately that

$$\text{DSIC} \implies \text{BIC}.$$

The converse fails: BIC only requires truth-telling to be optimal *on average* over the others' types, assuming they also tell the truth.

Interim notation. In the single-object IPV model, for a direct mechanism (q, t) define the **interim** allocation, payment, and utility of buyer i:

$$Q_i(\theta_i) = \mathbb{E}_{\theta_{-i}}\big[q_i(\theta_i, \theta_{-i})\big], \qquad T_i(\theta_i) = \mathbb{E}_{\theta_{-i}}\big[t_i(\theta_i, \theta_{-i})\big], \qquad U_i(\theta_i) = \theta_i Q_i(\theta_i) - T_i(\theta_i).$$

Since buyer i's utility is linear in (q_i, t_i), BIC reads simply

$$U_i(\theta_i) \;\geq\; \theta_i Q_i(\theta_i') - T_i(\theta_i') \qquad \text{for all } \theta_i, \theta_i'. \tag{1.3}$$

These interim objects are all that matter for incentives under BIC.

1.2.4 Individual Rationality

A designer usually cannot force agents to participate. Let $\bar{u}_i(\theta_i)$ denote the agent's **outside option**; in auctions $\bar{u}_i \equiv 0$.

Definition 1.10 (Individual rationality). A direct mechanism f is

(a) **ex post IR** if $u_i(f(\theta), \theta_i) \geq \bar{u}_i(\theta_i)$ for all i and all θ;

(b) **interim IR** if $\mathbb{E}_{\theta_{-i}}[u_i(f(\theta), \theta_i) \mid \theta_i] \geq \bar{u}_i(\theta_i)$ for all i and θ_i;

(c) **ex ante IR** if $\mathbb{E}_{\theta}[u_i(f(\theta), \theta_i)] \geq \mathbb{E}[\bar{u}_i(\theta_i)]$ for all i.

The three notions correspond to the moment at which an agent can walk away: after everything is known, after learning one's own type, or before learning anything. Clearly ex post IR $\Rightarrow$ interim IR $\Rightarrow$ ex ante IR.

1.2.5 Feasibility

Feasibility collects the physical and financial constraints on outcomes.

- **Allocation feasibility**: $k(\theta) \in K$ for all θ; for instance $q_i \geq 0$, $\sum_i q_i \leq 1$ in a single-object auction.
- **Budget constraints** on payments. A mechanism runs **no deficit** if $\sum_i t_i(\theta) \geq 0$ for all θ, and is **(ex post) budget balanced** if $\sum_i t_i(\theta) = 0$ for all θ. Weaker versions require these only in expectation.

For a seller, budget balance is not a concern; she wants $\sum_i t_i$ to be large. For a society organizing trade among its members with no outside source of money, budget balance is essential, and we will see that it can conflict sharply with efficiency (Section 1.3.3).

1.3 Fundamental Results

1.3.1 The Revelation Principle

There are infinitely many conceivable mechanisms: auctions with any bidding rules, multi-round negotiations, lotteries, menus. The following result makes the design problem tractable.

Theorem 1.11 (Revelation Principle; Gibbard, 1973; Myerson, 1979). *Let $\Gamma = (M, g)$ be any mechanism and let $s^* = (s_1^*, \dots, s_n^*)$ be a Bayes–Nash equilibrium of the game it induces. Define the direct mechanism*

$$f(\theta) = g\big(s^*(\theta)\big) = g\big(s_1^*(\theta_1), \dots, s_n^*(\theta_n)\big).$$

Then f is BIC, and truthful reporting in f produces the same outcome as s^* in Γ in every state θ. If s^* is an equilibrium in dominant strategies, then f is DSIC.

Proof (following Myerson, 1979). Fix agent i and types θ_i, θ_i'. If i reports θ_i' in f while the others report truthfully, the outcome is $g(s_i^*(\theta_i'), s_{-i}^*(\theta_{-i}))$, the same as if, in Γ, type θ_i sent the message

$s_i^*(\theta_i')$ instead of $s_i^*(\theta_i)$. Since $s_i^*(\theta_i)$ is a best response for type θ_i in Γ against s_{-i}^*,

$$\begin{aligned}\mathbb{E}_{\theta_{-i}}\big[u_i(f(\theta_i,\theta_{-i}),\theta_i) \mid \theta_i\big] &= \mathbb{E}_{\theta_{-i}}\big[u_i(g(s_i^*(\theta_i),s_{-i}^*(\theta_{-i})),\theta_i) \mid \theta_i\big]\\ &\geq \mathbb{E}_{\theta_{-i}}\big[u_i(g(s_i^*(\theta_i'),s_{-i}^*(\theta_{-i})),\theta_i) \mid \theta_i\big]\\ &= \mathbb{E}_{\theta_{-i}}\big[u_i(f(\theta_i',\theta_{-i}),\theta_i) \mid \theta_i\big],\end{aligned}$$

which is (1.2). If $s_i^*(\theta_i)$ is optimal against *every* profile of messages of the others, the same argument applies pointwise in θ_{-i} and gives (1.1). □

Interpretation. The direct mechanism "plays the equilibrium on the agents' behalf." Agents report types; the mechanism computes the messages they would have sent and implements the corresponding outcome. Lying in f is equivalent to pretending to be a different type in Γ, which was already unprofitable.

Remark. The revelation principle says that *for the purpose of finding what outcomes are achievable*, we may restrict attention to direct, incentive-compatible mechanisms. It does not say that direct mechanisms are better in practice. Indirect mechanisms may be simpler to explain, may require less communication, or may have fewer undesirable equilibria. The revelation principle only guarantees that truth-telling is *an* equilibrium of f, not the unique one.

From now on, the design problem becomes:

choose f to maximize the designer's objective, subject to IC, IR, and feasibility.

Myerson's own statement. In Myerson (1981) the revelation principle appears as Lemma 1: for any feasible auction mechanism there is an equivalent feasible direct revelation mechanism that gives the seller and all bidders the same expected utilities. Myerson's proof is exactly the construction above: the seller asks each bidder for her type, computes the strategy she would have used in the original auction, and implements the corresponding outcome.

The generalized revelation principle. In many problems agents not only *know* something the principal does not (adverse selection) but also *do* something the principal cannot control (moral hazard): a manager reports demand conditions and then chooses effort. Myerson (1982) extends the revelation principle to cover both at once.

Definition 1.12 (Generalized principal–agent problem; Myerson, 1982). There is a principal (agent 0) and agents $i = 1,\dots,n$. Each agent i has a finite set of types T_i and a finite set of **private decisions** D_i that the principal cannot observe or control; the principal has decisions D_0. Types are drawn from P on $T = T_1 \times \cdots \times T_n$, and player i has utility $U_i(d,t)$, $d \in D = D_0 \times D_1 \times \cdots \times D_n$.
A **direct coordination mechanism** is a map $\pi(\cdot \mid t) \in \Delta(D)$: each agent reports a type, and then the principal chooses d_0 and privately *recommends* a decision d_i to each agent i. Agent i is **honest and obedient** if she reports her true type and follows the recommendation. The mechanism is **incentive compatible** if honesty and obedience form a Bayesian equilibrium: for all i, all $t_i, \tau_i \in T_i$ and all $\delta_i : D_i \to D_i$,

$$\sum_{t_{-i}}\sum_{d} P(t)\,\pi(d \mid t)\,U_i(d,t) \;\geq\; \sum_{t_{-i}}\sum_{d} P(t)\,\pi(d \mid t_{-i},\tau_i)\,U_i\big((d_{-i},\delta_i(d_i)),t\big), \tag{1.4}$$

where both sums are over profiles with i's true type equal to t_i.

Theorem 1.13 (Generalized revelation principle; Myerson, 1982). *Given any coordination mechanism (with arbitrary, possibly multistage, message spaces) and any Bayesian equilibrium of it, there is an incentive-compatible direct coordination mechanism that gives the principal, and every agent, the same expected utility when agents are honest and obedient. Moreover, when all T_i and D_i are finite, finding the principal's optimal mechanism is a **linear program**: maximize $\sum_t \sum_d P(t)\pi(d \mid t)U_0(d,t)$ over π subject to* (1.4) *and $\pi(\cdot \mid t) \in \Delta(D)$.*

Proof (following Myerson, 1982, Propositions 1–2). Fix a mechanism and an equilibrium in which agent i uses reporting strategy ρ_i and decision rule $\delta_i(m_i, t_i)$ after receiving message m_i. Define $\pi^*(d \mid t)$ as the probability that, when types are t and everyone follows the equilibrium, the principal chooses d_0 and each agent ends up choosing d_i. Then π^* produces the same joint distribution of (t, d), hence the same expected utilities. If some type τ_i could gain in π^* by reporting $\hat{\tau}_i$ and then applying $\delta : D_i \to D_i$ to the recommendations, then in the original mechanism type τ_i could gain by reporting as $\hat{\tau}_i$ would and then applying δ to the decisions $\hat{\tau}_i$ would have taken. This contradicts equilibrium. The constraints (1.4) and the objective are linear in π, and with finite sets there are finitely many deviations (τ_i, δ_i), which gives the linear program. □

Interpretation. There are two incentive constraints: **honesty** (do not lie about what you know) and **obedience** (do not deviate from what you are told to do). Two special cases are already in these notes.

- With no private decisions ($|D_i| = 1$), only honesty remains: this is Theorem 1.11.
- With no private information ($|T_i| = 1$) and $|D_0| = 1$, only obedience remains, and the incentive-compatible mechanisms are exactly Aumann's **correlated equilibria**. Adding a payoff-relevant state that the principal knows gives the Bayes correlated equilibria of Chapter 2 (Section 2.3).

So Myerson (1982) is already a unified theory of mechanism design and information design, as Chapter 3 discusses.

1.3.2 Revenue Equivalence

The next result characterizes incentive compatibility in the single-object IPV model. It is the workhorse of the whole theory.

Lemma 1.14 (Myerson, 1981). *In the single-object IPV model, a direct mechanism (q, t) is BIC if and only if for every buyer i:*

(a) (monotonicity) *Q_i is nondecreasing; and*

(b) (envelope formula) *for all $\theta_i \in [\underline{\theta}, \bar{\theta}]$,*

$$U_i(\theta_i) = U_i(\underline{\theta}) + \int_{\underline{\theta}}^{\theta_i} Q_i(s)\, ds. \tag{1.5}$$

Proof (following Myerson, 1981; see also Krishna, 2009, Ch. 5). Drop the index i. Note that

$$\theta Q(\theta') - T(\theta') = U(\theta') + (\theta - \theta')Q(\theta'),$$

so (1.3) is equivalent to

$$U(\theta) \geq U(\theta') + (\theta - \theta')\, Q(\theta') \qquad \text{for all } \theta, \theta'. \tag{1.6}$$

($\Rightarrow$) Apply (1.6) twice, with the roles of θ and θ' exchanged:

$$(\theta - \theta')\, Q(\theta') \;\leq\; U(\theta) - U(\theta') \;\leq\; (\theta - \theta')\, Q(\theta).$$

If $\theta > \theta'$, dividing by $\theta - \theta'$ gives $Q(\theta') \leq Q(\theta)$, which is (a). The same inequalities, together with $0 \leq Q \leq 1$, give $|U(\theta) - U(\theta')| \leq |\theta - \theta'|$, so U is Lipschitz and hence absolutely continuous. Dividing by $\theta - \theta'$ and letting $\theta' \to \theta$ shows that $U'(\theta) = Q(\theta)$ at every point where Q is continuous, which is all but countably many points because Q is monotone. Integrating gives (b).

($\Leftarrow$) Suppose (a) and (b) hold and take any θ, θ'. By (b),

$$U(\theta) - U(\theta') - (\theta - \theta')Q(\theta') = \int_{\theta'}^{\theta} \big(Q(s) - Q(\theta')\big)\, ds.$$

If $\theta > \theta'$, the integrand is nonnegative by (a). If $\theta < \theta'$, the integral equals $\int_{\theta}^{\theta'} (Q(\theta') - Q(s))\, ds$, whose integrand is again nonnegative. Either way (1.6) holds. □

Interpretation. Incentive compatibility pins down utilities, and therefore payments, from the allocation rule alone, up to a constant. A higher type can always imitate a lower one, so she must be left an **information rent** equal to the area under Q_i; the rent grows faster where she is more likely to receive the good.

Since U_i is nondecreasing (its derivative Q_i is nonnegative), interim IR reduces to a single constraint on the lowest type:

$$U_i(\theta_i) \geq 0 \text{ for all } \theta_i \qquad \Longleftrightarrow \qquad U_i(\underline{\theta}) \geq 0.$$

Theorem 1.15 (Revenue Equivalence; Myerson, 1981; Riley and Samuelson, 1981). *In the single-object IPV model, consider two BIC mechanisms with the same interim allocation rules $Q_1, \ldots, Q_n$ and the same utilities $U_1(\underline{\theta}), \ldots, U_n(\underline{\theta})$ for the lowest types. Then every type of every buyer has the same expected payment in the two mechanisms. In particular, they raise the same expected revenue.*

Proof (following Myerson, 1981). By (1.5), $U_i(\theta_i)$ is the same in both mechanisms. Hence $T_i(\theta_i) = \theta_i Q_i(\theta_i) - U_i(\theta_i)$ is the same too. Expected revenue is $\sum_i \mathbb{E}[T_i(\theta_i)]$. □

Example 1.16 (First-price versus second-price). Two buyers have values drawn independently from the uniform distribution on $[0, 1]$.
Second-price auction. Bidding one's value is dominant (Proposition 1.8). The winner pays the lower of the two values, so expected revenue is

$$\mathbb{E}[\min(\theta_1, \theta_2)] = \int_0^1 \mathbb{P}(\min > x)\, dx = \int_0^1 (1 - x)^2\, dx = \tfrac{1}{3}.$$

First-price auction. The symmetric equilibrium is $b(\theta) = \theta/2$. (Check: if the rival bids $\theta_j/2$, bidding b wins with probability $\mathbb{P}(\theta_j < 2b) = 2b$, so the payoff is $(\theta - b)\, 2b$, maximized at

$b = \theta/2$.) The winner is the buyer with the higher value and pays half of it, so expected revenue is

$$\tfrac{1}{2}\,\mathbb{E}[\max(\theta_1, \theta_2)] = \tfrac{1}{2} \cdot \tfrac{2}{3} = \tfrac{1}{3}.$$

Both auctions give the good to the highest-value buyer and give zero utility to type 0, so Theorem 1.15 predicts equal revenue, as we have found.

The same argument works pointwise in θ_{-i} for DSIC mechanisms.

Corollary 1.17 (DSIC payment formula). *In the single-object private values model, (q, t) is DSIC if and only if for every i and θ_{-i}, the map $\theta_i \mapsto q_i(\theta_i, \theta_{-i})$ is nondecreasing and*

$$t_i(\theta_i, \theta_{-i}) = t_i(\underline{\theta}, \theta_{-i}) + \theta_i\, q_i(\theta_i, \theta_{-i}) - \underline{\theta}\, q_i(\underline{\theta}, \theta_{-i}) - \int_{\underline{\theta}}^{\theta_i} q_i(s, \theta_{-i})\, ds.$$

For a deterministic rule in which i wins exactly when θ_i exceeds a threshold $\tau_i(\theta_{-i})$, and the lowest type pays 0, this formula says the winner pays her threshold: the smallest report with which she would still have won. The second-price auction is the case $\tau_i(\theta_{-i}) = \max_{j \neq i} \theta_j$.

1.3.3 Limitations of Incentive-Compatible Mechanisms

Incentive compatibility is a real constraint. Three classical results show how much it can cost.

Without money: Gibbard–Satterthwaite. Suppose there are no transfers, X is finite, and each agent's type is a strict preference ranking over X, any ranking being possible. A social choice function is **dictatorial** if there is an agent i such that $f(\theta)$ is always i's top-ranked outcome.

Theorem 1.18 (Gibbard, 1973; Satterthwaite, 1975). *If $|X| \geq 3$ and f is onto X and DSIC on the unrestricted domain of strict preferences, then f is dictatorial.*

We do not prove this result here; see Börgers (2015), Ch. 8. It explains why the quasilinear model, where money can be used to align incentives, is so central, and why non-dictatorial strategy-proof rules (such as median voting with single-peaked preferences) require a restricted domain.

With money but a balanced budget: Myerson–Satterthwaite. A seller owns one good and has cost c; a buyer has value v. Trade is efficient exactly when $v \geq c$. Let $c \sim F_s$ and $v \sim F_b$ be independent with positive densities on a common interval $[\underline{\theta}, \bar{\theta}]$. A mechanism specifies the probability of trade $q(v, c)$, the buyer's payment $t_b(v, c)$, and the seller's receipt $t_s(v, c)$.

Theorem 1.19 (Myerson and Satterthwaite, 1983). *In this bilateral trade model there is no mechanism that is simultaneously BIC, interim IR for both traders, ex post efficient ($q = 1$ if $v > c$ and $q = 0$ if $v < c$), and runs no expected deficit ($\mathbb{E}[t_b - t_s] \geq 0$).*

Proof (following Myerson and Satterthwaite, 1983; Krishna, 2009, Ch. 5). Let $S = \mathbb{E}[(v - c)^+] > 0$ be the expected gains from trade under the efficient rule. The buyer trades with interim probability $Q_b(v) = \mathbb{P}(c \leq v) = F_s(v)$ and the seller with $Q_s(c) = \mathbb{P}(v \geq c) = 1 - F_b(c)$.

Applying Lemma 1.14 to the buyer, and to the seller (whose utility $t_s - cq$ is *decreasing* in her type, so the binding IR constraint is at $c = \bar{\theta}$), we get

$$U_b(v) = U_b(\underline{\theta}) + \int_{\underline{\theta}}^{v} F_s(s)\,ds, \qquad U_s(c) = U_s(\bar{\theta}) + \int_{c}^{\bar{\theta}} \big(1 - F_b(s)\big)\,ds.$$

With IR, $U_b(\underline{\theta}) \geq 0$ and $U_s(\bar{\theta}) \geq 0$. Now

$$\int_{\underline{\theta}}^{v} F_s(s)\,ds = \mathbb{E}_c\left[\int_{\underline{\theta}}^{v} \mathbf{1}\{c \leq s\}\,ds\right] = \mathbb{E}_c\big[(v-c)^+\big],$$

and similarly $\int_c^{\bar{\theta}} (1 - F_b(s))\,ds = \mathbb{E}_v[(v-c)^+]$. Taking expectations,

$$\mathbb{E}[U_b(v)] \geq S \qquad \text{and} \qquad \mathbb{E}[U_s(c)] \geq S.$$

But the expected budget surplus is total surplus minus what the traders keep:

$$\mathbb{E}[t_b - t_s] = \mathbb{E}[(v-c)\,q] - \mathbb{E}[U_b] - \mathbb{E}[U_s] \leq S - 2S = -S < 0. \qquad \square$$

Interpretation. Each trader must be given an information rent, and in the efficient mechanism each rent is as large as the entire expected surplus. Only an outside subsidy can pay for both. This is a fundamental reason why bargaining under private information breaks down even when gains from trade are certain to exist (here, whenever $v > c$).

Example 1.20 (A discrete Myerson–Satterthwaite problem; Bierbrauer and Netzer). The buyer's value is $\underline{\theta}_b = 20$ or $\bar{\theta}_b = 100$, and the seller's cost is $\underline{\theta}_s = 0$ or $\bar{\theta}_s = 80$, all independent and equally likely. Trade is efficient except when $(\theta_b, \theta_s) = (20, 80)$. Consider the "fair" rule f^*: trade efficiently at the price $(\theta_b + \theta_s)/2$, so that the two traders split the gains equally and every type gets a nonnegative payoff.
f^* is not BIC. A high-value buyer who tells the truth trades at price 50 (if the cost is 0) or 90 (if it is 80), and expects $\frac{1}{2}(50) + \frac{1}{2}(10) = 30$. If she claims a value of 20, she pays only 10 when the cost is 0 and does not trade otherwise, expecting $\frac{1}{2}(90) + \frac{1}{2}(0) = 45 > 30$.
Bierbrauer and Netzer (2012) note that these numbers give a discrete version of Theorem 1.19: no efficient mechanism is BIC, interim IR, and budget balanced. Solving the corresponding linear program (my own calculation) shows that any efficient, BIC and interim IR mechanism needs an expected subsidy of at least 5.

Efficiency and budget balance under DSIC. In rich quasilinear domains, the only efficient DSIC mechanisms are the Groves mechanisms of Section 1.4.4 (Green and Laffont, 1977; Holmström, 1979a), and in general no Groves mechanism is budget balanced (Green and Laffont, 1979). By contrast, if one weakens DSIC to BIC and types are independent, the expected-externality mechanism of d'Aspremont and Gérard-Varet (1979) and Arrow (1979) is efficient and exactly budget balanced; what it cannot always guarantee is interim IR, which is precisely the Myerson–Satterthwaite obstacle.

1.3.4 When Does the Revelation Principle Fail?

The revelation principle is a theorem, so it "fails" only when one of its assumptions fails. Reading the proof of Theorem 1.11 carefully, it uses three things:

(i) the designer **commits** to the outcome function;

(ii) agents care only about the **outcome** x and their type, $u_i(x, \theta_i)$, and not about which message they sent or which messages were available;

(iii) we only need truth-telling to be *an* equilibrium, not the unique one.

Each assumption can fail. Limited commitment (Bester and Strausz, 2001) and competition among several designers (Epstein and Peters, 1999) break (i). Multiple equilibria (iii) are the subject of implementation theory. Here we look at two papers about (ii).

Intentions: Bierbrauer and Netzer. Bierbrauer and Netzer (2012) let agents have *intention-based social preferences* in the style of Rabin (1993). Agent i's utility is her material payoff plus a term $y_{ij}\,\kappa_{ij}\,\kappa_{ji}$ for each other agent j, where κ_{ij} measures how kind i is to j and $y_{ij} \geq 0$ is the weight on reciprocity. Kindness is measured relative to an *equitable payoff*: the average of the best and worst (Pareto-efficient) payoffs i could have given j. The equilibrium notion is a **Bayes–Nash fairness equilibrium** (BNFE). It reduces to BNE when all $y_{ij} = 0$.

The crucial feature is that kindness depends on *what else i could have done*. So adding messages that are never used in equilibrium can change behavior, which violates assumption (ii).

Example 1.21 (Unused messages make fair trade possible; Bierbrauer and Netzer). Return to Example 1.20. In the direct mechanism for f^*, truth-telling is never a BNFE, for any reciprocity weights. In a truthful equilibrium each trader gets exactly her equitable payoff, so all kindness terms are zero, and the traders behave selfishly, which brings us back to the BNE analysis above.
Now enlarge the message sets. The buyer may also send $\underline{\theta}_b$ ("even lower than 20"), which works like reporting 20 but gives her an extra discount $\delta_b > 0$ when trade occurs. The seller may send $\bar{\bar{\theta}}_s$, which works like reporting 80 but raises the price by $\delta_s > 0$. Neither message is used in equilibrium. But because each trader *could* have grabbed the extra money and did not, telling the truth is now *kind*: in the truthful profile the buyer's kindness toward the seller is $\delta_b/4$, and the seller's is $\delta_s/4$. Choosing $\delta_b = 4/y_s$ and $\delta_s = 4/y_b$ turns each trader's utility into total material surplus. Since f^* is efficient, truth-telling is then a BNFE.

Theorem 1.22 (Bierbrauer and Netzer, 2012). *In the independent private values model with intention-based preferences (all $y_{ij} > 0$) and a commonly known kindness process:*

(a) The revelation principle fails: some social choice functions are implementable in BNFE but not by a truthful direct mechanism.

*(b) An **augmented revelation principle** holds: it is without loss to use mechanisms whose message sets* contain *the type sets and that have a truthful equilibrium (Mookherjee and Reichelstein, 1990).*

(c) Every materially efficient social choice function is implementable in BNFE, and the construction can also satisfy voluntary participation relative to any inefficient status quo.

Interpretation. Part (c) turns the Myerson–Satterthwaite impossibility into a possibility. The construction resembles a Groves mechanism (Section 1.4.4): it aligns each agent's objective with total welfare. The difference is that Groves does this with payments agents *make*; Bierbrauer and Netzer do it with payments agents *refuse to take*.

Robustness through insurance. The construction above requires the designer to know the reciprocity weights y. Without that knowledge, Bierbrauer and Netzer (2012) propose to shut the reciprocity channel down instead.

Definition 1.23 (Insurance property; Bierbrauer and Netzer). A social choice function has the **insurance property** if, for all $i \neq j$, agent i's expected payoff (averaging over everyone but j) does not depend on j's type. Equivalently, in the direct mechanism, no agent can change another agent's expected payoff by a unilateral misreport.

If j cannot affect i's payoff, kindness and unkindness have no outlet, so agents behave as selfish maximizers. Truth-telling then remains an equilibrium for *every* reciprocity weight y (a "psychologically robust" equilibrium).

Theorem 1.24 (Bierbrauer and Netzer, 2012). *Let f be implementable in BNE in the independent private values model. Then there is a social choice function $\tilde{f}$ with the same allocation rule, the same expected total transfers, and the same interim payoffs for every type of every agent, that is implementable in BNE* and *has the insurance property.*

Proof idea (following Bierbrauer and Netzer, 2012). Let $\rho_i(\theta_i)$ be type θ_i's interim payoff under f. Redefine i's transfer so that *ex post* her payoff equals $\rho_i(\theta_i)$ for every θ_{-i}. Her expected transfer is unchanged, so interim payoffs and incentives are unchanged, and her payoff no longer depends on the others' types. □

The price of insurance is that budget balance may then hold only in expectation, not ex post. For example, there is an insured version of the second-price auction with the same interim payoffs.

A claimed failure: Wu (2020). Wu (2020) argues that the revelation principle fails whenever agents' strategies are *real actions* (paying money, exerting effort) rather than abstract messages. His argument runs through Myerson's honest-and-obedient direct mechanism (Theorem 1.13): an agent could report a false type and then simply ignore the recommendation and take the action her true type would take. This produces the same outcome as honesty and obedience. Wu then assumes (his Note 6) that an agent who is indifferent prefers to lie in order to protect her privacy, and concludes that "dishonest and disobedient" is the equilibrium.

Remark (My assessment)**.** I do not think this argument shows that the revelation principle is false, for two reasons.

1. The revelation principle claims only that honesty and obedience is *an* equilibrium of the direct mechanism, by assumption (iii). Showing that another strategy is equally good shows that equilibrium is not unique, which is already acknowledged in the remark after Theorem 1.11.

2. The privacy assumption changes the agents' preferences: utility now depends on the report itself, not only on the outcome. This violates assumption (ii), so it is not a counterexample within the standard model; it is a new model, closer in spirit to Bierbrauer and Netzer, where preferences also depend on more than outcomes. Within the standard model, Myerson's proof (Theorem 1.13) explicitly allows agents to take private actions.

The paper is an unrefereed working paper. It is nevertheless useful as a prompt: it asks what happens when agents have a preference over *what they reveal*, which is exactly the kind of departure that could make lying optimal (Appendix A.1).

1.4 Optimal Mechanism Design

1.4.1 The Principal's Optimization Problem

We return to the seller of Section 1.1, now facing n buyers in the IPV model. By the revelation principle she chooses a direct mechanism (q, t) to solve

$$\max_{q,t} \ \mathbb{E}\Big[\sum_{i=1}^{n} t_i(\theta)\Big] \quad \text{subject to} \quad \text{BIC}, \ \text{interim IR}, \ q_i \geq 0, \ \textstyle\sum_i q_i \leq 1.$$

The key step is to express revenue in terms of the allocation rule alone.

Definition 1.25 (Virtual value). The **virtual value** of buyer i with type θ_i is

$$\varphi_i(\theta_i) = \theta_i - \frac{1 - F_i(\theta_i)}{f_i(\theta_i)}.$$

The distribution F_i is **regular** if φ_i is nondecreasing.

Lemma 1.26 (Myerson, 1981). *For any BIC mechanism,*

$$\mathbb{E}\big[t_i(\theta)\big] = \mathbb{E}\big[\varphi_i(\theta_i)\, q_i(\theta)\big] - U_i(\underline{\theta}). \tag{1.7}$$

Proof (following Myerson, 1981). Since $T_i = \theta_i Q_i - U_i$, it suffices to compute $\mathbb{E}[U_i(\theta_i)]$. By (1.5) and Fubini's theorem,

$$\begin{aligned} \mathbb{E}[U_i(\theta_i)] - U_i(\underline{\theta}) &= \int_{\underline{\theta}}^{\bar{\theta}} \int_{\underline{\theta}}^{\theta_i} Q_i(s)\, ds\, f_i(\theta_i)\, d\theta_i = \int_{\underline{\theta}}^{\bar{\theta}} Q_i(s) \int_{s}^{\bar{\theta}} f_i(\theta_i)\, d\theta_i\, ds \\ &= \int_{\underline{\theta}}^{\bar{\theta}} Q_i(s)\,\big(1 - F_i(s)\big)\, ds = \mathbb{E}\left[Q_i(\theta_i)\, \frac{1 - F_i(\theta_i)}{f_i(\theta_i)}\right]. \end{aligned}$$

Therefore $\mathbb{E}[T_i] = \mathbb{E}\big[(\theta_i - \frac{1-F_i(\theta_i)}{f_i(\theta_i)}) Q_i(\theta_i)\big] - U_i(\underline{\theta}) = \mathbb{E}[\varphi_i(\theta_i)\, q_i(\theta)] - U_i(\underline{\theta})$, using $Q_i(\theta_i) = \mathbb{E}[q_i(\theta) \mid \theta_i]$. □

Interpretation. Selling to type θ_i yields its value θ_i, minus the information rent that must then be paid to all higher types to stop them from imitating θ_i. The virtual value is the *marginal revenue* of serving type θ_i; Bulow and Roberts (1989) make this analogy with the monopolist's marginal revenue curve precise.

1.4.2 Optimal Auction Design

By (1.7), expected revenue is

$$\mathbb{E}\Big[\sum_i \varphi_i(\theta_i)\, q_i(\theta)\Big] - \sum_i U_i(\underline{\theta}).$$

The seller should set $U_i(\underline{\theta}) = 0$ (the smallest value allowed by IR), and then maximize the expected sum of virtual values *pointwise*: for each θ, give the good to the buyer with the highest virtual value, provided it is nonnegative.

Theorem 1.27 (Optimal auction; Myerson, 1981). *Suppose every F_i is regular. Then the following mechanism maximizes expected revenue among all BIC and interim IR mechanisms:*

$$q_i^*(\theta) = \begin{cases} 1 & \text{if } \varphi_i(\theta_i) > \max\{0,\ \max_{j\neq i} \varphi_j(\theta_j)\}, \\ 0 & \text{otherwise,} \end{cases}$$

with payments given by Corollary 1.17 and $t_i(\underline{\theta}, \theta_{-i}) = 0$. The winner pays the smallest type with which she would still have won. This mechanism is in fact DSIC and ex post IR.

Proof (following Myerson, 1981). For any BIC and IR mechanism, (1.7) and $U_i(\underline{\theta}) \geq 0$ give

$$\text{Revenue} \leq \mathbb{E}\Big[\sum_i \varphi_i(\theta_i)\, q_i(\theta)\Big] \leq \mathbb{E}\Big[\max\{0, \max_i \varphi_i(\theta_i)\}\Big],$$

since $q_i \geq 0$ and $\sum_i q_i \leq 1$. The rule q^* attains the second bound pointwise. By regularity, $q_i^*(\theta_i, \theta_{-i})$ is nondecreasing in θ_i for every θ_{-i}, so by Corollary 1.17 the stated payments make q^* DSIC (hence BIC). The lowest type pays 0 and wins with probability 0 (outside measure-zero ties), so $U_i(\underline{\theta}) = 0$ and the first bound also holds with equality. Ex post IR holds because a winner pays a threshold no larger than her own type. □

Corollary 1.28 (Symmetric case). *If all buyers draw values from the same regular distribution F, the optimal auction is a **second-price auction with reserve price** $r^* = \varphi^{-1}(0)$. The reserve does not depend on the number of buyers.*

Proof. With a common increasing φ, the highest virtual value belongs to the highest type, and it is nonnegative exactly when that type is at least r^*. The winner's threshold is $\max\{r^*, \max_{j\neq i} \theta_j\}$. □

Example 1.29 (Uniform values). Let F be uniform on $[0,1]$, so $F(\theta) = \theta$, $f(\theta) = 1$ and

$$\varphi(\theta) = \theta - (1-\theta) = 2\theta - 1, \qquad r^* = \tfrac{1}{2}.$$

(a) *One buyer.* The optimal mechanism is a posted price of $1/2$, with revenue $1/4$. This confirms the claim of Example 1.1: no mechanism, however complicated, beats the simple posted price.

(b) *Two buyers.* The optimal auction is a second-price auction with reserve $1/2$. Revenue equals the expected maximum of the virtual values, cut at 0. With $m = \max(\theta_1, \theta_2)$, which has density $2m$ on $[0,1]$,

$$\mathbb{E}\big[(2m-1)^+\big] = \int_{1/2}^{1} (2m-1)\, 2m\, dm = \Big[\tfrac{4}{3}m^3 - m^2\Big]_{1/2}^{1} = \tfrac{5}{12}.$$

This exceeds the revenue $1/3$ of the second-price auction without a reserve (Example 1.16).

Asymmetric bidders. When bidders differ, the optimal auction is *not* efficient even among the bidders who are served. Myerson (1981) gives the following example.

Example 1.30 (Discriminating among bidders; Myerson, 1981). Let θ_i be uniform on $[0, b_i]$. Then $\varphi_i(\theta_i) = \theta_i - (b_i - \theta_i) = 2\theta_i - b_i$, so the seller favors the bidder whose value distribution is *lower*. Take $b_1 = 1$, $b_2 = 2$, and seller value 0:

- bidder 1 has reserve $1/2$ and bidder 2 has reserve 1;
- bidder 1 beats bidder 2 whenever $2\theta_1 - 1 > 2\theta_2 - 2$, i.e. $\theta_1 > \theta_2 - \tfrac{1}{2}$.

With $\theta = (0.8, 1.1)$, bidder 1 wins although $\theta_1 < \theta_2$, and pays her threshold $\max\{\tfrac{1}{2}, \theta_2 - \tfrac{1}{2}\} = 0.6$. Handicapping the strong bidder forces her to bid more aggressively when her value is high.

Myerson's more general model. Myerson (1981) allows two features we have suppressed. First, the seller has her own value t_0 for the object (we took $t_0 = 0$); she sells only if the highest virtual value exceeds t_0. Second, values may have a common component through **revision effects**: if bidder i learned all the estimates t, she would revise her value to $v_i(t) = t_i + \sum_{j \neq i} e_j(t_j)$. The virtual value becomes $c_i(t_i) = t_i - e_i(t_i) - (1 - F_i(t_i))/f_i(t_i)$, and the analysis is otherwise the same.

The irregular case: ironing.

Theorem 1.31 (Ironing; Myerson, 1981). *Without regularity, for each bidder define, on quantiles $q \in [0,1]$,*

$$h_i(q) = c_i\big(F_i^{-1}(q)\big), \qquad H_i(q) = \int_0^q h_i(r)\, dr,$$

*let G_i be the **convex hull** of H_i (the largest convex function below it), and let $\bar{c}_i(t_i) = G_i'(F_i(t_i))$ be the **ironed virtual value**. Then an optimal auction gives the object to a bidder with the highest $\bar{c}_i(t_i)$, provided it exceeds t_0 (splitting ties equally), with payments $x_i(t) = p_i(t)\, v_i(t) - \int_{a_i}^{t_i} p_i(t_{-i}, s_i)\, ds_i$.*

Since G_i is convex, $\bar{c}_i$ is nondecreasing, so the allocation is monotone and Lemma 1.14 applies. Where H_i lies strictly above G_i, $\bar{c}_i$ is flat. Types there are **pooled**: they win with the same probability. In the regular case $G_i = H_i$ and nothing changes.

Correlated values. Independence matters a great deal. Myerson (1981), Section 7, shows what can happen without it.

Example 1.32 (Full surplus extraction; Myerson, 1981). Two bidders have values 10 or 100, with

$$\mathbb{P}(10,10) = \mathbb{P}(100,100) = \tfrac{1}{3}, \qquad \mathbb{P}(10,100) = \mathbb{P}(100,10) = \tfrac{1}{6}.$$

Consider the mechanism: if both report 100, sell to one at random for 100; if one reports 100 and the other 10, sell to the high one for 100 *and charge the low one* 30; if both report 10, give the object to one at random and pay each bidder 10 on average.
A bidder with value 10 thinks the other has 100 with probability $1/3$, so her "side bet" (pay 30 if the other is high, net gain 15 if the other is low) is worth $\tfrac{1}{3}(-30) + \tfrac{2}{3}(15) = 0$. A bidder

with value 100 thinks the other is high with probability 2/3. If she pretends to have value 10, she pays 30 when the other is high, and when the other is low she gets the object (worth 100 to her) half the time plus 10 on average, so she expects $\frac{2}{3}(-30) + \frac{1}{3}(50 + 10) = 0$, no more than the 0 she gets by telling the truth. So truth-telling is an equilibrium, the object always goes to a highest-value bidder, every type gets zero, and expected revenue is

$$\tfrac{1}{3}(100) + \tfrac{1}{6}(130) + \tfrac{1}{6}(130) + \tfrac{1}{3}(-20) = 70,$$

the entire expected surplus. If the seller may never pay bidders, the best she can do is $66\frac{2}{3}$: keep the object when both report 10, and otherwise sell to a high bidder for 100. (I verified both numbers by solving the linear programs.)

This is the precursor of the Crémer–McLean theorem (Theorem 3.1). Correlation lets the seller cross-check each report against the others, and information rents disappear.

1.4.3 Optimal Mechanisms in Quantile Space

Bergemann et al. (2026) recast the optimal mechanism problem in a way that makes ironing look like concavification, the same tool as in Chapter 2. The trick is to describe everything by **quantile functions**.

Setting. A seller faces a unit mass of buyers with unit demand and utility $v \cdot q - p$, where q is the quality of the good received (or the probability of receiving it). Buyers are indexed by their quantile $t \in [0, 1]$ in the value distribution. The seller has an exogenous *inventory* of goods whose qualities have distribution F_q. Write

$$V(t) = F_v^{-1}(t), \qquad Q(t) = F_q^{-1}(t)$$

for the quantile value function and the quantile inventory. A mechanism assigns quantile t an expected quality $X(t)$, nondecreasing in t. If buyers are imperfectly informed, replace V by the quantile function W of their expected values.

Auctions as a special case. With N symmetric bidders and one object, a bidder at quantile t wins an efficient auction with probability t^{N-1}. Setting $Q(t) = t^{N-1}$ turns the screening problem into the symmetric auction problem, with revenue measured per bidder.

Lemma 1.33 (Revenue in quantile space; Bergemann, Heumann and Morris, 2026). *For an incentive-compatible mechanism that gives the lowest type zero surplus,*

$$R(W, X) = X(0)\, W(0) + \int_0^1 (1 - t)\, W(t)\, dX(t).$$

Proof (following Bergemann, Heumann and Morris, 2026). By the envelope theorem (Lemma 1.14 in quantile form), the surplus of quantile t is $u(t) = \int_0^t X(s)\, dW(s)$, so the payment is $p(t) = W(t)X(t) - \int_0^t X(s)\, dW(s)$. Quantiles are uniform on $[0, 1]$; integrating $p(t)$ over t and integrating by parts twice gives the formula. □

Revenue is *bilinear* in (W, X): linear in the allocation for fixed information, and linear in the information for a fixed allocation.

Feasibility as majorization. The seller can pool goods (offer lotteries over qualities) and throw goods away, but cannot create quality. Following Kleiner et al. (2021), an allocation X is feasible if and only if

$$\int_x^1 X(t)\,dt \le \int_x^1 Q(t)\,dt \qquad \text{for all } x \in [0,1],$$

written $X \prec_w Q$ (X is **weakly majorized** by Q). In the auction case this is Border's condition. Importantly, this constraint does not involve the value distribution.

Theorem 1.34 (Optimal mechanism via concavification; Bergemann, Heumann and Morris, 2026). *Let $r_W(t) = W(t)(1-t)$ be the **pointwise revenue function** (the revenue from a posted price $W(t)$ that sells to the top $1-t$ of buyers), let t_m maximize r_W, and let $\bar{r}_W$ be its concavification. An optimal mechanism excludes all quantiles $t < t_m$ (a reserve price $W(t_m)$); above t_m, it gives $X(t) = Q(t)$ where $\bar{r}_W = r_W$, and on each interval where $\bar{r}_W > r_W$ it gives every quantile the average of Q over that interval (pooling). Optimal revenue is*

$$\bar{r}_W(t_m)\,Q(t_m) + \int_{t_m}^1 \bar{r}_W(t)\,dQ(t).$$

Connection to Myerson. Differentiating the pointwise revenue function,

$$-r_W'(t) = W(t) - W'(t)(1-t) = \varphi(t),$$

which is the virtual value written in quantiles (with $W = V$, $V'(t) = 1/f(v)$ and $1-t = 1-F(v)$). So Myerson's $H(q) = \int_0^q \varphi = V(0) - r_V(q)$. Taking the convex hull of H is the same as taking the concave hull of r_V: **Myerson's ironing is concavification of the revenue curve**. Regularity (φ increasing) is the same as r_V being concave, in which case there is no pooling (Corollary 1.28).

Example 1.35 (Checking the uniform case). Uniform values give $V(t) = t$ and $r_V(t) = t(1-t)$, which is concave with $t_m = 1/2$. With two bidders, $Q(t) = t$, and revenue per bidder is

$$r_V(\tfrac{1}{2})\,Q(\tfrac{1}{2}) + \int_{1/2}^1 t(1-t)\,dt = \tfrac{1}{8} + \tfrac{1}{12} = \tfrac{5}{24}.$$

Two bidders give total revenue $5/12$, as in Example 1.29.
Bergemann et al. (2026) use $V(t) = Q(t) = t^4$ (five bidders) as a running example. Here $r_V(t) = t^4(1-t)$ lies strictly below its concavification on $[0, 3/4]$, but $t_m = 4/5$, so the region that would be ironed is excluded anyway. The optimal auction is still a second-price auction with reserve $V(4/5)$.

The same framework handles *information* design (choose W for fixed X) and *joint* design (choose both). We return to it in Section 3.2.

1.4.4 The Vickrey–Clarke–Groves (VCG) Mechanism

Now consider a designer who wants *efficiency* rather than revenue, in a general quasilinear environment with private values: allocations $k \in K$ and utilities $v_i(k, \theta_i) - t_i$. An allocation rule k^* is **efficient** if

$$k^*(\theta) \in \arg\max_{k \in K} \sum_{j=1}^n v_j(k, \theta_j) \qquad \text{for all } \theta.$$

Definition 1.36 (Groves mechanisms). A **Groves mechanism** uses an efficient allocation rule k^* and payments

$$t_i(\theta) = h_i(\theta_{-i}) - \sum_{j \neq i} v_j\big(k^*(\theta), \theta_j\big),$$

where h_i is any function that does not depend on θ_i. The **VCG** (or **Clarke pivot**) mechanism is the Groves mechanism with

$$h_i(\theta_{-i}) = \max_{k \in K} \sum_{j \neq i} v_j(k, \theta_j).$$

Theorem 1.37 (Vickrey, 1961; Clarke, 1971; Groves, 1973). *Every Groves mechanism is DSIC.*

Proof (following Groves, 1973). Fix i, θ_{-i}, and θ_i. If i reports θ_i', her utility is

$$v_i\big(k^*(\theta_i', \theta_{-i}), \theta_i\big) + \sum_{j \neq i} v_j\big(k^*(\theta_i', \theta_{-i}), \theta_j\big) - h_i(\theta_{-i}).$$

The last term does not depend on her report. The first two terms are total welfare, evaluated at the true types, of the allocation $k^*(\theta_i', \theta_{-i})$. By definition of k^*, this is maximized by the allocation $k^*(\theta_i, \theta_{-i})$, which the truthful report produces. □

Interpretation. The Groves payment makes each agent's payoff equal to total welfare, up to a term she cannot affect. Every agent therefore "wants what the planner wants."

The Clarke pivot term has a natural meaning. Let $W(\theta) = \max_k \sum_j v_j(k, \theta_j)$ be maximal welfare and $W_{-i}(\theta_{-i})$ the maximal welfare of the others if i were absent. Under VCG,

$$t_i(\theta) = \underbrace{W_{-i}(\theta_{-i}) - \sum_{j \neq i} v_j\big(k^*(\theta), \theta_j\big)}_{\text{externality } i \text{ imposes on others}} \;\geq\; 0,$$

so payments are never negative, and i's utility is $W(\theta) - W_{-i}(\theta_{-i})$, her marginal contribution to welfare. If $v_i(k, \theta_i) \geq 0$ for all k, this is nonnegative, so VCG is ex post IR.

Example 1.38 (Two identical items). Two identical items are sold to three buyers who each want at most one item. Values are $\theta = (10, 7, 4)$. The efficient allocation gives items to buyers 1 and 2; $W = 17$.

- Buyer 1: without her, the best allocation serves buyers 2 and 3, so $W_{-1} = 11$. In the chosen allocation, the others get 7. She pays $11 - 7 = 4$.
- Buyer 2: $W_{-2} = 14$, the others get 10, so she pays 4.
- Buyer 3: she does not change the allocation, so she pays 0.

Each winner pays the highest losing value. With one item, the same computation gives the second-price auction.

1.4.5 Efficiency, Incentives, and Revenue Trade-offs

The optimal auction and the VCG mechanism answer different questions, and comparing them shows the basic trade-off of the theory.

Example 1.39 (The cost of the reserve). Two buyers, values uniform on $[0,1]$. Let $m = \max(\theta_1, \theta_2)$.

	Revenue	Total surplus	Lost surplus
Second-price (VCG)	$1/3$	$\mathbb{E}[m] = 2/3$	0
Optimal auction (reserve $1/2$)	$5/12$	$\int_{1/2}^{1} m \cdot 2m\, dm = 7/12$	$1/12$

The seller gains $5/12 - 1/3 = 1/12$ in revenue, buyers lose $1/3 - 1/6 = 1/6$ in utility, and $1/12$ of surplus is destroyed because the good sometimes goes unsold.

Some general lessons:

1. **Revenue versus efficiency.** A revenue-maximizing designer distorts the allocation downward (here, by withholding the good) to reduce information rents. Exactly the same logic gives quantity distortions for low types in monopoly screening (Mussa and Rosen, 1978).

2. **Competition versus design.** Bulow and Klemperer (1996) show that, with i.i.d. regular values, a second-price auction with $n+1$ buyers raises more revenue than the optimal auction with n buyers. With $n = 1$ and uniform values: $1/3 > 1/4$. Attracting one more bidder is worth more than perfect design.

3. **Efficiency versus budget balance.** Efficiency with DSIC needs Groves payments, which generally cannot balance the budget; efficiency with BIC and IR can require subsidies (Theorem 1.19).

4. **Simplicity versus optimality.** Optimal mechanisms depend on the prior through φ_i. VCG does not. When the designer is unsure of the prior, prior-free and robust mechanisms become attractive.

1.5 Further Questions and Research Directions

Question (1)**.** The revelation principle requires commitment. If the seller cannot commit not to renegotiate (for example, she can lower the price after a buyer refuses), what outcomes remain achievable? How does the answer depend on the number of periods? (A starting point is the literature on the Coase conjecture and on mechanism design with limited commitment.)

Question (2)**.** Revenue equivalence relies on independent private values and risk-neutral buyers. Which of the three assumptions (independence, private values, risk neutrality) fails in a setting you care about, and in which direction does the ranking of first- and second-price auctions change? Compare with Milgrom and Weber (1982).

Question (3)**.** In Theorem 1.19, the efficient mechanism runs a deficit of at least S. What is the *second-best* mechanism, the one that maximizes expected gains from trade subject to BIC, IR and budget balance? What does it look like when both traders' values are uniform on $[0,1]$?

Question (4)**.** Optimal mechanisms depend on the prior. How much revenue is lost by using a mechanism that performs well for *every* prior in some class? What is the right benchmark for comparison?

Question (5)**.** Bierbrauer and Netzer use unused messages to make truthful reporting *kind*. Could a seller in a single-object auction use the same device to raise revenue above Myerson's bound (Theorem 1.27) when bidders have intention-based preferences toward the seller? What would the equitable payoff be?

Question (6)**.** Myerson's generalized revelation principle (Theorem 1.13) reduces finite problems to linear programs. Formulate Example 1.20 as such a linear program, with trade probabilities and transfers as variables, and verify that the smallest expected subsidy is 5. Which constraints bind?

Question (7)**.** With multiple goods and additive values, the optimal mechanism is generally not a separate Myerson auction for each good, and may involve bundling or lotteries. Why does the one-dimensional argument of Section 1.4 break down?

CHAPTER 2

Information Design

2.1 Introduction and Motivation

2.1.1 Information as an Economic Instrument

In Chapter 1 the designer controlled the *rules* of the game (who gets what, who pays what) and took the agents' information as given. In many situations the opposite is true: the rules are fixed, but someone controls *what the players know*.

Examples

- a school chooses how coarse its grades are, which affects which students employers hire (Ostrovsky and Schwarz, 2010);
- a rating agency chooses how many rating categories to publish;
- a regulator chooses what results of bank stress tests to disclose;
- an online platform chooses what product information to show to consumers;
- a prosecutor chooses which investigations to carry out before a trial.

In each case the information provider cannot change the receivers' payoffs or force their actions. She can only shape their *beliefs*. Information design asks: what can be achieved this way, and how?

2.1.2 Motivating Example: Information Disclosure

The following example, due to Kamenica and Gentzkow (2011), runs through this chapter.

Example 2.1 (Prosecutor and judge). A defendant is either guilty (G) or innocent (I). Both the prosecutor and the judge start from the common prior

$$\mathbb{P}(G) = 0.3, \qquad \mathbb{P}(I) = 0.7.$$

The judge chooses to *convict* or *acquit*. She gets utility 1 from a correct decision and 0 otherwise, so she convicts if and only if her belief that the defendant is guilty is at least $1/2$. The prosecutor gets utility 1 from a conviction and 0 from an acquittal, regardless of guilt. The prosecutor cannot lie or hide evidence once found, but she chooses which investigations to conduct, and therefore what the judge will learn. Consider three options.

(a) *No investigation.* The judge's belief stays at $0.3 < 1/2$, so she always acquits. Conviction probability: 0.

(b) *Full investigation.* The judge learns the truth and convicts exactly the guilty. Conviction probability: 0.3.

(c) *A partially informative investigation.* The investigation always produces a "bad" report c when the defendant is guilty, and produces it with probability $3/7$ when the defendant is innocent. Otherwise it produces a "good" report a. Then

$$\mathbb{P}(c) = 0.3 + 0.7 \cdot \tfrac{3}{7} = 0.6, \qquad \mathbb{P}(G \mid c) = \frac{0.3}{0.6} = \tfrac{1}{2}, \qquad \mathbb{P}(G \mid a) = 0.$$

After c, the judge (just) prefers to convict; after a, she acquits. Conviction probability: 0.6.

The judge is perfectly rational and knows how the investigation is designed, yet the prosecutor doubles the conviction rate relative to full disclosure. We will show that 0.6 is the best she can do.

2.1.3 The Sender–Receiver Framework

Definition 2.2 (Persuasion problem). A **persuasion problem** consists of:

(i) a finite set of **states** Ω, with a **prior** $\mu_0 \in \Delta(\Omega)$ that has full support;

(ii) a **Receiver** with a compact set of actions A and a continuous utility $u(a, \omega)$;

(iii) a **Sender** with utility $v(a, \omega)$.

The Sender first chooses a **signal** (or **information structure**) π, which consists of a finite set of signal realizations S and a family of distributions $\pi(\cdot \mid \omega) \in \Delta(S)$, one for each state. Then the state ω is drawn from μ_0, a realization s is drawn from $\pi(\cdot \mid \omega)$, the Receiver observes π and s, and she chooses an action.

The key assumption, again, is **commitment**: the Sender chooses π *before* learning the state, and the Receiver knows π. This separates Bayesian persuasion from two older models of communication.

Model	What the sender controls	Credibility of messages
Cheap talk (Crawford and Sobel, 1982)	message after seeing ω	none; must be self-enforcing
Verifiable disclosure (Grossman, 1981; Milgrom, 1981)	which true facts to reveal	cannot lie
Bayesian persuasion (Kamenica and Gentzkow, 2011)	the signal π, before ω	full commitment

2.1.4 States of the World, Priors, and Beliefs

A **belief** is an element $\mu \in \Delta(\Omega)$. Given a belief μ, the Receiver chooses

$$\hat{a}(\mu) \in \arg\max_{a \in A} \; \mathbb{E}_{\mu}[u(a, \omega)] = \arg\max_{a \in A} \sum_{\omega} \mu(\omega)\, u(a, \omega),$$

breaking ties in favor of the Sender. The Sender's resulting expected payoff, as a function of the Receiver's belief, is

$$\hat{v}(\mu) = \mathbb{E}_{\mu}\big[v(\hat{a}(\mu), \omega)\big].$$

Everything about the Receiver's behavior that matters to the Sender is contained in $\hat{v}$. In Example 2.1, identifying a belief with $\mu = \mathbb{P}(G)$,

$$\hat{v}(\mu) = \begin{cases} 1, & \mu \geq 1/2, \\ 0, & \mu < 1/2. \end{cases}$$

Remark. The tie-breaking rule makes $\hat{v}$ upper semicontinuous, which guarantees that the optimal signal exists. It is the analog of the assumption in Chapter 1 that indifferent agents tell the truth.

2.2 Bayesian Persuasion

2.2.1 Information Structures and Signal Design

A signal π affects the Receiver only through the beliefs it induces. After realization s, the Receiver's **posterior** is given by Bayes' rule:

$$\mu_s(\omega) = \frac{\pi(s \mid \omega)\, \mu_0(\omega)}{\sum_{\omega'} \pi(s \mid \omega')\, \mu_0(\omega')}. \tag{2.1}$$

Since s is random, so is the posterior. A signal therefore induces a **distribution of posteriors** $\tau \in \Delta(\Delta(\Omega))$, where $\tau(\mu)$ is the probability that the posterior equals μ.

Example 2.3 (Distributions of posteriors). In Example 2.1, again writing $\mu = \mathbb{P}(G)$:

- no investigation induces $\tau = \delta_{0.3}$ (the posterior equals the prior);
- full investigation induces $\tau(1) = 0.3$, $\tau(0) = 0.7$;
- the partial investigation induces $\tau(1/2) = 0.6$, $\tau(0) = 0.4$.

In every case the posteriors average to the prior: $0.3 \cdot 1 + 0.7 \cdot 0 = 0.3$ and $0.6 \cdot \tfrac{1}{2} + 0.4 \cdot 0 = 0.3$.

2.2.2 Bayesian Updating and Posterior Beliefs

The observation at the end of Example 2.3 is general and, remarkably, it is the *only* restriction.

Definition 2.4 (Bayes plausibility). A distribution of posteriors τ with finite support is **Bayes plausible** if

$$\sum_{\mu \in \operatorname{supp} \tau} \tau(\mu)\, \mu = \mu_0.$$

Proposition 2.5 (Splitting lemma; Aumann and Maschler, 1995; Kamenica and Gentzkow, 2011). *A finite-support distribution of posteriors τ is induced by some signal if and only if it is*

Bayes plausible.

Proof (following Kamenica and Gentzkow, 2011). ($\Rightarrow$) Let π induce τ, and let $p(s) = \sum_\omega \pi(s \mid \omega)\mu_0(\omega)$ be the probability of realization s. By (2.1), $p(s)\,\mu_s(\omega) = \pi(s \mid \omega)\,\mu_0(\omega)$. Summing over s,

$$\sum_s p(s)\,\mu_s(\omega) = \mu_0(\omega) \sum_s \pi(s \mid \omega) = \mu_0(\omega).$$

Grouping realizations that lead to the same posterior gives $\sum_\mu \tau(\mu)\mu = \mu_0$.

($\Leftarrow$) Let τ be Bayes plausible. Take $S = \operatorname{supp}\tau$, so realizations are posteriors, and define

$$\pi(\mu \mid \omega) = \frac{\tau(\mu)\,\mu(\omega)}{\mu_0(\omega)}.$$

This is a probability distribution over S for each ω, because $\sum_\mu \tau(\mu)\mu(\omega) = \mu_0(\omega)$ by Bayes plausibility. The probability of realization μ is $\sum_\omega \mu_0(\omega)\pi(\mu \mid \omega) = \tau(\mu)\sum_\omega \mu(\omega) = \tau(\mu)$, and the posterior after it is $\mu_0(\omega)\pi(\mu \mid \omega)/\tau(\mu) = \mu(\omega)$. So π induces τ. □

Interpretation. The Sender can "split" the prior into any collection of posteriors that average back to it, with any weights that make the average work. She cannot move the *average* belief, because the Receiver knows how the signal was designed; but she has full control over how beliefs are spread around it.

2.2.3 The Sender's Optimization Problem

By Proposition 2.5, the Sender's problem of choosing a signal is equivalent to choosing a distribution of posteriors:

$$V(\mu_0) = \max_\tau \sum_\mu \tau(\mu)\,\hat{v}(\mu) \quad \text{subject to} \quad \sum_\mu \tau(\mu)\,\mu = \mu_0. \tag{2.2}$$

This reformulation removes the Receiver entirely: her rationality is summarized in $\hat{v}$, and Bayesian updating in the single linear constraint.

Definition 2.6 (Concave closure). The **concave closure** of $\hat{v}$ is the smallest concave function that lies above it:

$$(\operatorname{cav}\hat{v})(\mu) = \sup\Big\{ \sum_k \lambda_k \hat{v}(\mu_k) \;:\; \lambda_k \geq 0,\ \sum_k \lambda_k = 1,\ \sum_k \lambda_k \mu_k = \mu \Big\}.$$

Theorem 2.7 (Concavification; Aumann and Maschler, 1995; Kamenica and Gentzkow, 2011). *The value of the Sender's problem is*

$$V(\mu_0) = (\operatorname{cav}\hat{v})(\mu_0).$$

The Sender benefits from persuasion, that is, $V(\mu_0) > \hat{v}(\mu_0)$, if and only if $\hat{v}(\mu_0) < (\operatorname{cav}\hat{v})(\mu_0)$. An optimal signal needs at most $|\Omega|$ realizations.

Proof (following Kamenica and Gentzkow, 2011). The feasible set in (2.2) consists exactly of the weights λ_k and points μ_k appearing in the definition of cav $\hat{v}$ at μ_0, so the two suprema coincide. Upper semicontinuity of $\hat{v}$ ensures the supremum is attained. Geometrically, $(\mu_0, V(\mu_0))$ lies on the boundary of the convex hull of the graph of $\hat{v}$, a subset of $\mathbb{R}^{|\Omega|-1} \times \mathbb{R}$ (beliefs live in a simplex of dimension $|\Omega| - 1$); by Carathéodory's theorem it is a convex combination of at most $|\Omega|$ points of the graph. The second claim is the comparison of $V(\mu_0)$ with the value $\hat{v}(\mu_0)$ of the uninformative signal. □

Two useful special cases follow immediately.

- If $\hat{v}$ is **concave**, then cav $\hat{v} = \hat{v}$ and *no disclosure* is optimal.
- If $\hat{v}$ is **convex**, then full disclosure is optimal: cav $\hat{v}$ is the linear function that equals $\hat{v}$ at the degenerate beliefs δ_ω, and full disclosure splits μ_0 exactly into those beliefs.

2.2.4 Optimal Information Disclosure

Example 2.8 (Prosecutor and judge, solved). With $\mu = \mathbb{P}(G)$, $\hat{v}(\mu) = \mathbf{1}\{\mu \geq 1/2\}$. Its concave closure is

$$(\text{cav}\,\hat{v})(\mu) = \min\{2\mu,\, 1\},$$

shown in Figure 2.1. At the prior, $V(0.3) = 0.6$. The optimal split puts the posterior at the two kinks, $\mu = 0$ and $\mu = 1/2$, with weights solving $\tau(1/2) \cdot \frac{1}{2} = 0.3$, so $\tau(1/2) = 0.6$. This is exactly signal (c) of Example 2.1.

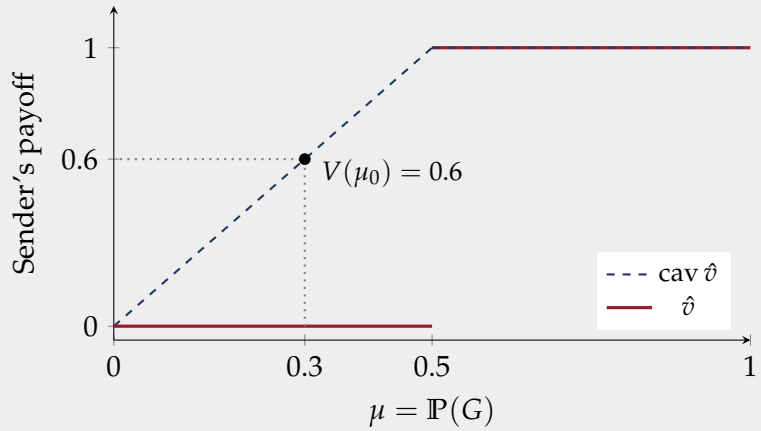


Figure 2.1: The Sender's value function $\hat{v}$ (solid) and its concave closure (dashed).

Straightforward signals. The optimal signal in the example has only two realizations, one for each action, and the judge follows the "recommendation" contained in each. This is general.

Proposition 2.9 (Kamenica and Gentzkow, 2011). *Without loss of generality, the Sender can use a **straightforward** signal: the set of realizations is a subset of A, and the Receiver's optimal action after realization a is a itself.*

Proof (following Kamenica and Gentzkow, 2011). Take any signal and merge all realizations after which the Receiver takes the same action a into one realization labelled a. The posterior after

the merged realization is a weighted average of the posteriors after the original realizations. Since a maximizes $\mathbb{E}_\mu[u(\cdot,\omega)]$ at each of those posteriors, and this expectation is linear in μ, a also maximizes it at their average. So the Receiver still takes a, and the joint distribution of states and actions, hence the Sender's payoff, is unchanged. □

Interpretation. This is the **revelation principle of information design**. The Sender only needs to send action recommendations, and the incentive constraint becomes **obedience**: the Receiver must want to follow the recommendation. Section 2.3 develops this idea for many receivers.

Remark. Two features of the optimal signal in Example 2.8 are typical.

1. **The Receiver is left indifferent** after the "good" recommendation (posterior exactly $1/2$). If the judge strictly preferred to convict after c, the prosecutor could add more innocent defendants to c and still obtain a conviction.

2. **Bad news is fully revealing.** When the Sender's preferred action is not taken, the Receiver is told the worst: here, that the defendant is certainly innocent. Pooling the "bad" realization with anything else would waste probability that could be used to persuade.

These are the information-design analogs of "no distortion at the top" and of binding IR constraints for the lowest type in Chapter 1.

2.3 Bayes Correlated Equilibrium (BCE)

2.3.1 Correlated Recommendations and Information

We now let information affect several players who interact strategically.

Definition 2.10 (Basic game and information structure). A **basic game** G consists of a finite set of states Θ with prior $\psi \in \Delta(\Theta)$, and for each player $i \in N$ a finite set of actions A_i and a utility $u_i : A \times \Theta \to \mathbb{R}$, where $A = \prod_i A_i$.
An **information structure** $\mathcal{S}$ consists of finite sets of signals T_i and a map $\pi : \Theta \to \Delta(T)$, where $T = \prod_i T_i$. Player i observes t_i.

The pair $(G, \mathcal{S})$ is a standard Bayesian game, and has Bayes–Nash equilibria $\beta_i : T_i \to \Delta(A_i)$. An equilibrium induces a joint distribution over states and actions. The information designer wants to know: as $\mathcal{S}$ varies, which distributions over (θ, a) can arise? Following Bergemann and Morris (2016), we take as given whatever information $\mathcal{S}$ the players already have, and let the designer add more.

2.3.2 Obedience Constraints

Definition 2.11 (Decision rule and obedience). A **decision rule** for $(G, \mathcal{S})$ is a map $\sigma : \Theta \times T \to \Delta(A)$. Think of a mediator who knows θ and t and privately recommends action a_i to each player i, with a drawn from $\sigma(\cdot \mid \theta, t)$.

σ is **obedient** if, for every i, every $t_i \in T_i$ and all $a_i, a_i' \in A_i$,

$$\sum_{\theta, t_{-i}, a_{-i}} \psi(\theta)\, \pi(t_i, t_{-i} \mid \theta)\, \sigma(a_i, a_{-i} \mid \theta, t) \Big[u_i(a_i, a_{-i}, \theta) - u_i(a_i', a_{-i}, \theta) \Big] \;\geq\; 0. \tag{2.3}$$

Interpretation. The left side is proportional to player i's expected gain from following the recommendation a_i rather than deviating to a_i', conditional on her signal t_i and the recommendation. Each player should want to obey, given what the recommendation reveals about the state and about others' recommendations.

2.3.3 The BCE Feasibility Problem

Definition 2.12 (Bayes correlated equilibrium; Bergemann and Morris, 2016). A decision rule σ is a **Bayes correlated equilibrium** (BCE) of $(G, \mathcal{S})$ if it is obedient.

For fixed $(G, \mathcal{S})$, (2.3) is a finite collection of *linear* inequalities in the unknowns $\sigma(a \mid \theta, t)$. Together with $\sigma \geq 0$ and $\sum_a \sigma(a \mid \theta, t) = 1$, the set of BCE is a convex polytope, and a designer with a linear objective, say $\mathbb{E}[w(a, \theta)]$, solves a **linear program**:

$$\max_{\sigma} \sum_{\theta, t, a} \psi(\theta) \pi(t \mid \theta) \sigma(a \mid \theta, t)\, w(a, \theta) \quad \text{s.t.} \quad \sigma \text{ obedient}, \;\; \sigma(\cdot \mid \theta, t) \in \Delta(A).$$

The reason BCE is the right object is the following equivalence.

Theorem 2.13 (Bergemann and Morris, 2016). *A decision rule σ is a BCE of $(G, \mathcal{S})$ if and only if there is an information structure $\mathcal{S}^*$ that expands $\mathcal{S}$ (each player observes her signal in $\mathcal{S}$ and possibly more) and a Bayes–Nash equilibrium of $(G, \mathcal{S}^*)$ that induces the same joint distribution of (θ, t, a) as σ.*

Proof (following Bergemann and Morris, 2016). ($\Rightarrow$) Let σ be obedient. Define $\mathcal{S}^*$ by letting each player observe (t_i, r_i), where $r = (r_1, \ldots, r_n)$ is drawn from $\sigma(\cdot \mid \theta, t)$. Let each player play $\beta_i(t_i, r_i) = r_i$. Conditional on $(t_i, r_i = a_i)$, inequality (2.3) says exactly that a_i is a best response when everyone else follows their r_j. So β is a BNE of $(G, \mathcal{S}^*)$, and by construction it induces σ.

($\Leftarrow$) Let $\mathcal{S}^*$ give each player (t_i, r_i), with extra signals r drawn from some $\lambda(\cdot \mid \theta, t)$, and let β be a BNE. The induced decision rule is

$$\sigma(a \mid \theta, t) = \sum_{r} \lambda(r \mid \theta, t) \prod_{j} \beta_j(a_j \mid t_j, r_j).$$

Fix i, t_i, a_i, a_i'. For each r_i, the BNE condition says that if $\beta_i(a_i \mid t_i, r_i) > 0$, then a_i is at least as good as a_i' for type (t_i, r_i). Multiplying that inequality by $\beta_i(a_i \mid t_i, r_i)$ and summing over r_i gives exactly (2.3), since each term is nonnegative. □

Interpretation. BCE describes *everything* that can happen in equilibrium when the players know at least $\mathcal{S}$, without having to list all the information structures. The recommendation-sending mediator plays the role of the direct mechanism in Theorem 1.11.

Example 2.14 (Prosecutor as a linear program). One player (the judge), $\Theta = \{G, I\}$, null initial information. Let $x = \sigma(\text{convict} \mid G)$ and $y = \sigma(\text{convict} \mid I)$. The obedience constraints are

$$\begin{aligned} \text{told "convict":} \quad & 0.3\,x \cdot (1-0) + 0.7\,y \cdot (0-1) \geq 0 \iff 0.3\,x \geq 0.7\,y, \\ \text{told "acquit":} \quad & 0.7\,(1-y) \geq 0.3\,(1-x). \end{aligned}$$

The prosecutor maximizes $0.3x + 0.7y$. The first constraint binds at the optimum, giving $x = 1$, $y = 3/7$, and value 0.6, as in Example 2.8.

2.3.4 Connections to Bayesian Games

Complete information: correlated equilibrium. If Θ is a single point and $\mathcal{S}$ is trivial, (2.3) reduces to Aumann's definition of correlated equilibrium (Aumann, 1974).

Example 2.15 (Chicken; Aumann, 1974). Two drivers choose C (chicken out) or D (dare), with payoffs

	C	D
C	$6,6$	$2,7$
D	$7,2$	$0,0$

In the symmetric mixed Nash equilibrium each player dares with probability $1/3$ (making the other indifferent: $6 - 4p = 7 - 7p$ at $p = 1/3$), and each earns $14/3$.
Now let a mediator draw one of (C,C), (C,D), (D,C) with probability $1/3$ each and privately tell each player her own action.

- A player told D knows the other was told C: D gives $7 > 6$.
- A player told C thinks the other was told C or D with probability $1/2$ each: C gives $\frac{1}{2}(6) + \frac{1}{2}(2) = 4$, while D gives $\frac{1}{2}(7) + \frac{1}{2}(0) = 3.5$.

So obedience holds, and each player earns $\frac{1}{3}(6+2+7) = 5 > 14/3$. Correlated information improves on every Nash equilibrium without changing the game.

More information shrinks the BCE set. Adding information to $\mathcal{S}$ means adding obedience constraints conditional on finer events, so the BCE set gets smaller. Bergemann and Morris (2016) show that this ordering characterizes informativeness: $\mathcal{S}$ is "individually sufficient" for $\mathcal{S}'$ (a many-player generalization of Blackwell's order) if and only if $\text{BCE}(G, \mathcal{S}) \subseteq \text{BCE}(G, \mathcal{S}')$ for every basic game G.

Information design in games. Putting the pieces together, an information designer who can choose any expansion of $\mathcal{S}$, and who can select her favorite equilibrium, solves the linear program of Section 2.3.3. The single-receiver persuasion problem of Section 2.2 is the special case $|N| = 1$, $\mathcal{S}$ trivial. Bergemann and Morris (2019) and Taneva (2019) develop this unified view. If instead the designer fears that players will coordinate on the *worst* equilibrium for her, the BCE approach no longer applies directly, and the problem becomes much harder (Mathevet et al., 2020).

2.4 Further Questions and Research Directions

Question (1)**.** Theorem 2.7 assumes the Sender commits to the signal. In practice, a prosecutor could abandon an investigation halfway. What can the Sender achieve with partial or no commitment, and how does this connect to cheap talk (Crawford and Sobel, 1982)?

Question (2)**.** Suppose the Receiver has private information about her own preferences (for example, the judge's conviction threshold is unknown to the prosecutor). Is it still without loss to send action recommendations? Compare Kolotilin et al. (2017).

Question (3)**.** With a continuum of states and a Sender who cares only about the posterior mean, the problem becomes one of choosing a mean-preserving contraction of the prior (Gentzkow and Kamenica, 2016; Dworczak and Martini, 2019). Work out the optimal signal when the state is uniform on $[0,1]$ and the Receiver takes action 1 if and only if the posterior mean exceeds $2/3$.

Question (4)**.** How should Example 2.15 change if the mediator also knows a payoff-relevant state? Construct a two-player, two-state game and compute the designer-optimal BCE by linear programming.

Question (5)**.** In the BCE approach the designer picks the equilibrium. Under adversarial equilibrium selection, are there simple sufficient conditions (for example, supermodularity of the basic game) under which the optimal information structure can still be found by a linear program?

CHAPTER 3

The Intersection of Mechanism and Information Design

3.1 Information Asymmetry and Strategic Interaction

Chapters 1 and 2 studied two ways of influencing strategic behavior.

	Mechanism design	Information design
Designer controls	rules: allocations and payments	information: signals
Taken as given	players' information	the game (actions, payoffs)
Incentive constraint	truth-telling (IC)	obedience
Participation	IR	usually none
Revelation principle	Theorem 1.11	Proposition 2.9, Theorem 2.13
Typical tool	envelope theorem, virtual values	concavification, linear programming

The two problems are formally close. Myerson (1982) considers a principal who both collects reports from agents and sends them recommendations. His revelation principle (Theorem 1.13) says that one may restrict attention to mechanisms in which agents are **honest** (they report their types truthfully) and **obedient** (they follow the recommended actions). Mechanism design imposes only honesty, information design only obedience; the general problem imposes both.

The two theories also share a basic lesson: *the structure of information determines what can be achieved*. A striking illustration comes from mechanism design itself.

Theorem 3.1 (Crémer and McLean, 1988). *Consider a single-object auction with finitely many types in which buyers' types are correlated, and suppose the conditional beliefs $\mathbb{P}(\theta_{-i} \mid \theta_i)$ of the different types of each buyer are linearly independent. Then the seller can design a mechanism that is BIC and interim IR and extracts the* entire *surplus: the good goes to the highest-value buyer, who is left with zero expected utility.*

The idea is that each buyer's type is "cross-checked" against the others' reports through side bets that are fair only for truthful types. Information rents, the central cost of Chapter 1, disappear. Since this conclusion is generally regarded as unrealistic, the result is usually read as a sign that standard models are too sensitive to the fine details of the common prior, which motivates robust approaches to design.

3.2 Information Disclosure and Incentive Compatibility

Return to the seller of Chapter 1, but now suppose buyers do *not* know their values. They learn about them only through information the seller provides (product descriptions, test

drives, inspection of a house). The seller therefore chooses both a selling mechanism and an information structure.

Proposition 3.2 (One buyer). *Suppose one buyer has value $v \geq 0$ drawn from F, and initially knows only F. If the seller chooses both the information the buyer receives and the selling mechanism, the optimal choice is to reveal* nothing *and post the price $\mathbb{E}[v]$. The seller then extracts the entire surplus $\mathbb{E}[v]$.*

Proof. Whatever the buyer learns, he will participate only if his expected utility is nonnegative, so expected revenue is at most the expected surplus $\mathbb{E}[v\,q] \leq \mathbb{E}[v]$. With no information, the buyer's expected value is $\mathbb{E}[v]$ and he accepts the price $\mathbb{E}[v]$ (with the usual tie-breaking), so this bound is attained. □

Interpretation. Information creates private information, and private information creates rents. With a single buyer, the seller has no reason to let the buyer learn anything.

With several buyers the conclusion changes, because information is also needed to allocate the good well.

Example 3.3 (The value of disclosure with n buyers). Let n buyers have values i.i.d. uniform on $[0,1]$, unknown to the buyers. Compare two extreme information policies.

- *No disclosure.* All buyers have expected value $1/2$, so the seller sells to anyone at price $1/2$. Revenue: $1/2$.
- *Full disclosure* followed by Myerson's optimal auction (reserve $1/2$). With $M = \max_i \theta_i$, which has density nm^{n-1},

$$\begin{aligned}\text{Revenue} = \mathbb{E}\left[(2M-1)^+\right] &= \int_{1/2}^{1} (2m-1)\, nm^{n-1}\, dm \\ &= \frac{2n}{n+1}\left(1 - 2^{-(n+1)}\right) - \left(1 - 2^{-n}\right).\end{aligned}$$

n	No disclosure	Full disclosure + optimal auction
1	0.500	0.250
2	0.500	0.417
3	0.500	0.531
$\to \infty$	0.500	$\to 1$

With three or more buyers, full disclosure beats no disclosure: the gain from selling to the buyer who values the good most outweighs the information rents that disclosure creates.

Neither extreme is optimal in general. Bergemann and Pesendorfer (2007) show that the seller's optimal information structure partitions each buyer's values into intervals, so that buyers learn something but not everything, and that the partitions are asymmetric across buyers even when buyers are ex ante identical. Eső and Szentes (2007) study the case where buyers already have some private information and the seller controls additional information: she optimally discloses all of it, and by selling access to it through a suitable menu (the "handicap auction"), she captures all the additional surplus it creates, so the buyers earn rents only on their original

private information.

The roles can also be reversed: if the *buyer* chooses what to learn about his value, anticipating the seller's price, the buyer-optimal information structure makes the seller indifferent among a range of prices, so that trade always occurs and the buyer captures a large share of the surplus (Roesler and Szentes, 2017).

A third designer is a regulator or platform that controls what the *seller* knows about buyers.

Theorem 3.4 (Bergemann, Brooks and Morris, 2015). *Consider a monopolist facing a market with a given distribution of values (finitely many values), and let π^* be the profit from the optimal uniform price and w^* the efficient total surplus. By segmenting the market (revealing to the monopolist information about buyers' values) and letting the monopolist price optimally in each segment, one can achieve* every *pair of consumer surplus CS and producer surplus PS such that*

$$CS \geq 0, \qquad PS \geq \pi^*, \qquad CS + PS \leq w^*.$$

Interpretation. Third-degree price discrimination can be fully efficient with all surplus to consumers, fully efficient with all surplus to the seller, or as bad as uniform pricing, and everything in between. The welfare effects of "more information for sellers" depend entirely on *which* information.

Screening the Receiver: Persuasion with Type Reporting

In Chapter 2 the Sender knew the Receiver's preferences. Suppose instead that the Receiver has a privately known *type* $k \in K$ (her preferences), drawn from a known distribution λ. Without communication, the Sender must use one signal for all types. Castiglioni et al. (2022) add a mechanism-design step: before the state is realized, the Sender commits to a **menu** $\{\varphi^k\}_{k \in K}$ of signals; the Receiver reports a type k, and then receives signals from φ^k.

Definition 3.5 (IC menu; Castiglioni, Marchesi and Gatti, 2022). A menu is **incentive compatible** if each type prefers her own signal to any other type's, when in each case she best-responds to the signals she receives. A signal is **direct** if its realizations are action recommendations, and **persuasive** if the type who reported truthfully wants to follow them.

Proposition 3.6 (Castiglioni, Marchesi and Gatti, 2022). *With a single Receiver, there is an optimal menu consisting of direct and persuasive signals. Consequently, the Sender's optimal menu can be computed by a linear program of polynomial size. Without type reporting, the Sender's problem is NP-hard to approximate within any constant factor (as shown in earlier work by the same authors).*

The constraints combine the two theories: **obedience** for the truthful type, as in Section 2.3, and **honesty** across types, as in Chapter 1. This is a special case of Myerson's honest-and-obedient framework (Theorem 1.13).

Example 3.7 (When reporting helps; my own example). Three equally likely states $\omega_1, \omega_2, \omega_3$. The Receiver chooses to invest or not; the Sender wants investment in ev-

ery state. There are two equally likely Receiver types, with payoff from investing (not investing gives 0):

	ω_1	ω_2	ω_3
Type A	$+1$	-2	-1
Type B	-2	$+1$	-1

One signal for both types. At any posterior μ, both types invest only if $\mu_1 - 2\mu_2 - \mu_3 \geq 0$ and $-2\mu_1 + \mu_2 - \mu_3 \geq 0$. Adding these gives $-\mu_1 - \mu_2 - 2\mu_3 \geq 0$, which is impossible. So at most one type invests after any signal realization, and the Sender gets at most $1/2$. She achieves $1/2$: send "A" in ω_1 and with probability $1/2$ in ω_3, and "B" otherwise; type A invests after "A" and type B after "B".
Separate optimal signals fail IC. Type A's own Kamenica–Gentzkow signal recommends investing in ω_1 and ω_3, giving her expected utility 0. But if A reports B, she receives B's signal ("invest" in ω_2, ω_3), and its "do not invest" realization reveals ω_1, so she earns $1/3 > 0$ by investing there. Type A would lie.
Optimal menu. Solving the linear program gives value $5/9 > 1/2$, with type A's signal recommending investment with probability 1 in ω_1 and $1/3$ in ω_2 and ω_3 (and symmetrically for B). After "invest", type A is exactly indifferent: $\frac{1}{3}\left(1 - 2 \cdot \frac{1}{3} - \frac{1}{3}\right) = 0$. Type reporting raises the Sender's payoff by about 11%.
In a two-state version of the prosecutor example with judges who differ only in their conviction threshold, the same linear program gives no gain from reporting. This is in line with Kolotilin et al. (2017), who show that with such threshold types, menus achieve nothing beyond a single signal.

An Integrated Framework: Bergemann, Heumann and Morris

Section 1.4.3 showed that revenue is bilinear in the buyers' (quantile) expected values W and the (quantile) allocation X: $R(W, X) = X(0)W(0) + \int_0^1 (1-t)W(t)\,dX(t)$. Bergemann et al. (2026) use this to treat three problems with one tool.

1. Mechanism design (choose X, fix $W = V$): maximize a linear functional of X subject to $X \prec_w Q$. Solved by concavifying $r_V(t) = V(t)(1-t)$ (Theorem 1.34).

2. Information design (choose W, fix an efficient allocation $X = Q$). By Blackwell's theorem, a distribution of expected values can be generated by some information structure if and only if it is a mean-preserving contraction of the value distribution, which in quantiles reads $W \prec V$ (majorization). Integrating by parts,

$$R(W, Q) = e_Q(0)\, W(0) + \int_0^1 e_Q(t)\, dW(t), \qquad e_Q(t) = \int_t^1 \big(Q(s) - Q(t)\big)\, ds \quad (t > 0),$$

where $e_Q(t)$ is the **excess quality** available above quantile t. The optimal information structure pools values exactly where e_Q lies below its concavification.

Proposition 3.8 (Optimal disclosure; Bergemann, Heumann and Morris, 2026). *With an efficient allocation, if the distribution of qualities has an increasing hazard rate, revealing nothing is*

optimal; if it has a decreasing hazard rate, full disclosure is optimal.

For a single-object auction with N bidders ($Q(t) = t^{N-1}$) neither condition holds, and the optimal structure is **upper censorship**: bidders learn their values below a threshold quantile and only that they are above it otherwise. This sustains competition at the top and reduces the winner's information rent. The optimal threshold depends only on N.

3. Joint design (choose $W \prec V$ and $X \prec_w Q$). The problem is now bilinear with two majorization constraints.

Theorem 3.9 (Optimality of pooling; Bergemann, Heumann and Morris, 2026). *There is an optimal pair (W^*, X^*) in which both are monotone partitional with the same partition of quantiles. Moreover, every optimal pair consists of pooling intervals only: when the seller controls both information and allocation, no interval of buyers is fully informed and fully separated.*

Interpretation. Suppose an interval $[t_1, t_2]$ of buyers were fully informed and fully separated. Pooling the qualities on that interval costs only a third-order amount (bounded by the product of the standard deviations of virtual values and qualities on the interval, times its length). Pooling the *values* there as well raises the price of the quality step for *every* buyer above t_1, a second-order gain on a non-negligible mass of buyers. So pooling always pays.

This is a sharp version of the trade-off in Example 3.3: the seller wants buyers informed enough to sort them, but never fully informed.

3.3 Mechanism Design with Endogenous Information

In Chapter 1, agents' types were exogenous. In practice, agents often decide how much to learn before participating (inspecting a used car, evaluating a takeover target), and the mechanism affects the value of learning.

- **Information acquisition.** Bergemann and Välimäki (2002) show that in private-values environments, the VCG mechanism gives each agent socially efficient incentives to acquire information before reporting, because under VCG each agent's payoff is her marginal contribution to welfare (Section 1.4.4). With interdependent values, this alignment generally breaks down.

- **Selling information.** Information can itself be the good. A seller of information (a data provider, a consultant) faces a buyer with private beliefs and designs a menu of experiments and prices (Bergemann et al., 2018). Here mechanism design (screening the buyer's type) and information design (the experiments) are a single problem.

- **Information and contracts.** In moral hazard problems, the principal conditions pay on signals of the agent's effort. By the informativeness principle (Holmström, 1979b), any signal that carries information about effort should enter the optimal contract. Choosing a monitoring technology or a performance rating system is therefore an information design problem nested in a contracting problem.

3.4 Applications to Auctions, Contracting, and Market Design

Auctions. Classical auction theory fixes the information structure (e.g. IPV, or affiliated values in Milgrom and Weber, 1982) and compares formats. A newer approach fixes the format and asks what can happen across *all* information structures. For example, Bergemann et al. (2017) characterize the lowest revenue a first-price auction can generate across all information structures consistent with a given distribution of values. This gives robust revenue guarantees that do not rely on knowing what bidders know.

Contracting. Beyond monitoring, organizations design what employees learn about one another and about the firm's prospects (for instance, whether to publish relative performance). Each such choice changes incentives to exert effort and to stay, combining information design with the screening and moral hazard tools of contract theory (Laffont and Martimort, 2002).

Market design. Platforms continuously choose what to show: which sellers are visible, what ratings mean, how coarse school grades are (Ostrovsky and Schwarz, 2010). Theorem 3.4 shows that even a simple monopoly market can produce any division of surplus depending on the information released, so disclosure policy is a first-order design instrument alongside pricing and matching rules.

3.5 Open Questions and Potential Research Directions

Question (1)**.** Joint design: a seller who chooses both an information structure and an auction for n buyers with uniform values. Example 3.3 compared only the two extreme information structures. Characterize the optimal joint design for $n = 2$, and compare it with Bergemann and Pesendorfer (2007).

Question (2)**.** Under Theorem 3.1, correlation allows full surplus extraction. If the seller could also choose what buyers learn about one another, could she always create the correlation needed? What would limit her?

Question (3)**.** Both revelation principles depend on commitment. What happens when the designer can commit to the mechanism but not to the information she releases, or vice versa?

Question (4)**.** Can the honest-and-obedient framework of Myerson (1982) be combined with the linear-programming approach of BCE to give a computationally tractable theory of joint mechanism-and-information design for small finite games? What is the size of the resulting linear program?

Question (5)**.** Theorem 3.4 describes what is possible. Which segmentations would a consumer-protection regulator choose, and which information would she need to implement them?

CHAPTER 4

Conclusion

4.1 Main Results and Economic Implications

These notes have followed a single question through two lenses: how can a designer achieve her goals when the relevant information is dispersed among self-interested players?

1. **Revelation principles reduce design to constrained optimization.** Any equilibrium outcome of any mechanism can be achieved by a direct, truthful mechanism (Theorem 1.11); any equilibrium outcome under any information structure can be achieved by obedient recommendations (Proposition 2.9, Theorem 2.13). The designer chooses an outcome rule subject to incentive constraints, instead of searching over all possible games or signals.

2. **Private information has a price.** In mechanism design, incentive compatibility pins down payments from allocations (Lemma 1.14) and forces the designer to leave information rents. This yields revenue equivalence, Myerson's optimal auction with its virtual values and reserve prices, and the impossibility of efficient bilateral trade (Theorem 1.19).

3. **Efficiency is achievable, at a cost.** VCG mechanisms make truth-telling dominant for any efficient allocation problem, but they generally do not balance the budget and do not maximize revenue.

4. **Beliefs can be moved only on average.** In information design, the only constraint on the distribution of posteriors is Bayes plausibility (Proposition 2.5), and the Sender's value is the concave closure of her indirect payoff (Theorem 2.7). With many receivers, the set of achievable outcomes is the BCE polytope, and optimal design is a linear program.

5. **The two tools interact.** Information determines rents, and rents determine whether a designer wants to release information (Proposition 3.2, Example 3.3). Who controls information can change the division of surplus across the whole feasible range (Theorem 3.4).

4.2 Limitations and Extensions

- **Commitment.** Every result above assumes the designer can commit to a mechanism or a signal. Limited commitment (renegotiation, dynamic pricing, cheap talk) is a major direction of current research.

- **Common prior and detail-dependence.** Optimal mechanisms and signals depend on the prior, sometimes sharply (Theorem 3.1). Robust mechanism design and robust information design ask for guarantees that hold across many priors or information structures.

- **Equilibrium selection.** Revelation principles guarantee that truthful or obedient behavior is *an* equilibrium, not the only one. Designers who worry about unfavorable equilibria need stronger notions such as unique or full implementation.
- **Multidimensional types.** The clean one-dimensional theory of Chapter 1 does not extend directly to agents with several private characteristics, such as values for several goods.
- **Behavioral considerations.** Real agents may not be fully Bayesian or fully strategic. Simple mechanisms, such as those that are *obviously* strategy-proof (Li, 2017), are one response.

APPENDIX **A**

Working Projects

This appendix collects research projects I am currently developing. Each states the question, explains why it is not trivially answered by the results in the main text, and sets up a first model to work with. The appendix records work in progress, so statements here are conjectures or illustrative calculations unless explicitly proved.

A.1 When Is Lying Uniquely Optimal?

A.1.1 The question

Can there be environments in which, even when the seller chooses the buyer's information optimally,

(i) misreporting is the buyer's *unique* best response, and

(ii) the buyer's misreporting *raises* the seller's revenue compared with any outcome in which the buyer reports truthfully?

A.1.2 Why this is not trivial

The revelation principle (Theorem 1.11) seems to rule this out. If the buyer lies in equilibrium, the seller can build a new direct mechanism that "lies on the buyer's behalf" and obtain the same outcome with truth-telling. So if the seller can **commit** to **any** mechanism, lying can never be strictly better for her than every truthful outcome.

The project therefore has to drop at least one assumption behind the revelation principle. Candidate settings:

1. **Restricted formats.** The selling rule is fixed (for example, "pay what you report" or a first-price auction) and the seller controls only information. Bid shading in a first-price auction is a form of misreporting; the question is when information design makes it more valuable to the seller.

2. **Limited commitment.** The seller cannot commit not to use the report against the buyer later (for example, in repeated sales). A buyer who understates her value may be the only equilibrium, and the seller may prefer that equilibrium to one in which she learns the truth.

3. **Two designers.** The information designer and the seller are different players (a platform and a merchant). Lying by the buyer may benefit the merchant but not the platform, or the reverse.

4. **Multiple equilibria.** Truth-telling is *an* equilibrium of a direct mechanism, not necessarily the unique one. Could information design make a lying equilibrium the unique one?

5. **Preferences over reports.** If agents care about what they reveal, not only about outcomes, the revelation principle need not hold (Section 1.3.4). A preference for privacy, as assumed by Wu (2020), makes a lie strictly better whenever it leads to the same outcome; intention-based preferences (Bierbrauer and Netzer, 2012) make unused messages matter. Either could deliver "lying is uniquely optimal."

A.1.3 A first model to try

One buyer with value $v \in \{L, H\}$, $0 < L < H$, prior $\mathbb{P}(H) = \mu_0$. The seller chooses a signal π about v that the buyer observes, as in Chapter 2. The selling format is *fixed*: the buyer sends a report $r \in \{L, H\}$ and the seller then charges the price that is optimal given her belief after r, without having committed to prices in advance (setting 2 above).

Question (A.1)**.** In this model, characterize the information structures π under which the equilibrium has type H reporting L with positive probability. Is there a π for which this is the unique equilibrium? Compare the seller's revenue with the full-commitment benchmark of Proposition 3.2.

Question (A.2)**.** Does the answer change in setting 1, with the format fixed at a first-price auction with two buyers whose information the seller designs?

Related literature to read. Commitment and the revelation principle: Myerson (1982); Bester and Strausz (2001); preferences beyond outcomes: Bierbrauer and Netzer (2012); Wu (2020); information design in auctions: Bergemann and Pesendorfer (2007); Bergemann et al. (2017); equilibrium selection in information design: Mathevet et al. (2020).

A.2 Information Design as a Branching Process

A.2.1 The question

In Chapter 2 a signal reaches one receiver, or a fixed set of receivers. In many settings, information *spreads*: a persuaded receiver passes the message on, and the people she reaches may pass it on again. Rumors, product recommendations, and political messages travel this way. The process of who receives the message resembles a Galton–Watson branching process, in which each persuaded receiver produces a random number of new receivers.

How should a sender design information when its reach depends on the receivers' reactions?

A.2.2 A first model

Return to the prosecutor and judge (Example 2.1), but replace the single judge by a population of receivers. The state $\omega \in \{G, I\}$ is drawn once, with $\mathbb{P}(G) = 0.3$. The sender commits to a signal that recommends "act" with probability p_ω in state ω. The spreading rule is:

- each reached receiver sees an independent draw of the signal (conditionally independent given ω);

- a receiver who is told "act" and obeys forwards the message to a Poisson(λ) number of new receivers;
- a receiver who is told "do not act" forwards nothing.

Conditional on ω, each reached receiver acts with probability p_ω and then has Poisson(λ) children, so the number of children of a reached receiver who pass the message on has a Poisson(λp_ω) distribution. The set of acting receivers therefore forms a Galton–Watson tree with mean offspring λp_ω. Using the extinction results for branching processes (generating functions, as in MATH 425), the message dies out with probability q_ω, the smallest solution in $[0, 1]$ of

$$q = e^{\lambda p_\omega (q-1)},$$

and it spreads forever with positive probability if and only if $\lambda p_\omega > 1$.

Obedience. If each receiver updates only on her own signal, as the judge did in Chapter 2, obedience after "act" is the same constraint as before. With $p_G = 1$, it reads $p_I \leq 3/7$.

Example A.1 (Spread under the prosecutor's optimal signal). Take the prosecutor-optimal signal $p_G = 1$, $p_I = 3/7$ and $\lambda = 2$.

- If the defendant is guilty, the mean offspring is 2, and solving $q = e^{2(q-1)}$ gives $q_G \approx 0.203$: the message spreads indefinitely with probability about 0.80.
- If the defendant is innocent, the mean offspring is $2 \cdot 3/7 = 6/7 < 1$, so the message dies out with probability 1.

More generally, since obedience caps p_I at $3/7$, a message about an innocent defendant can spread forever only if $\lambda \cdot 3/7 > 1$, that is, $\lambda > 7/3$. Under these assumptions, the network's connectivity, not just the signal, decides whether persuasion "goes viral" in the bad state.

A.2.3 Open issues

Question (A.3)**. Selection on being reached.** A rational receiver should also update on the fact that the message reached her at all: being reached is more likely in the state where p_ω is higher. How does this change the obedience constraint deeper in the tree, and does it make persuasion easier or harder as the message spreads?

Question (A.4)**. The sender's objective.** If the sender values the total number of receivers who act, or the probability that the message survives, rather than the action of the first receiver, what is the optimal (p_G, p_I)? Is it still optimal to leave the receivers indifferent?

Question (A.5)**. Designed branching.** Could the sender design not only the signal but also the forwarding rule (for example, how many people a persuaded receiver may share with)? This would combine information design with a mechanism-design choice of the offspring distribution.

Related literature to read. Bayesian persuasion: Kamenica and Gentzkow (2011); information design with many receivers: Bergemann and Morris (2019); Taneva (2019). A systematic search for papers on persuasion and information diffusion in networks is still to be done.

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